Machine Learning

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1 k-Nearest Neighbors

1.1 The k-NN algorithm

1-NN. Let $\mathcal{D} = \{(\boldsymbol{x}_i, y_i)\}_{i=1}^N$ be a training set with features $\boldsymbol{x}_i \in \mathbb{R}^D$ and class labels $y_i \in \{1, \dots, C\}$. The **1-NN** rule fixes a distance measure d (e.g. Euclidean) and gives a new observation the label of its nearest training point; the same rule does regression by copying a real-valued y_i. 1-NN corresponds to the Voronoi tessellation of the training points: each point labels its own cell, and the decision region of a class is the union of the cells of its points. Every single point, a mislabelled one included, owns a cell, so 1-NN generalises poorly.

k-NN. Letting several neighbours vote and taking the majority makes the rule robust against errors in the training set.

Definition 1.1 (k-NN classifier)**.** Let $\mathcal{N}_k(\boldsymbol{x})$ be (the indices of) the k training points nearest to $\boldsymbol{x}$, and let the **indicator variable** $\mathbb{I}(e)$ be 1 if e is true and 0 otherwise. The **k-NN classifier** predicts

$$p(y = c \mid \boldsymbol{x}, k) = \frac{1}{k} \sum_{i \in \mathcal{N}_k(\boldsymbol{x})} \mathbb{I}(y_i = c), \qquad \hat{y} = \arg\max_c \; p(y = c \mid \boldsymbol{x}, k),$$

i.e. $\boldsymbol{x}$ gets the mode of its neighbours' labels.

For fixed $\boldsymbol{x}$, $p(\cdot \mid \boldsymbol{x}, k)$ is a probability mass function on the classes: it is nonnegative and, since each neighbour has exactly one label ($\sum_{c=1}^C \mathbb{I}(y_i = c) = 1$),

$$\sum_{c=1}^{C} p(y = c \mid \boldsymbol{x}, k) = \frac{1}{k} \sum_{i \in \mathcal{N}_k(\boldsymbol{x})} \sum_{c=1}^{C} \mathbb{I}(y_i = c) = \frac{k}{k} = 1.$$

It is the fraction of each class among the k neighbours, like the fraction of each colour in a jar: with 4 red, 10 green and 6 blue balls, $p(\text{red}) = 4/20 = 0.2$, $p(\text{green}) = 10/20 = 0.5$, $p(\text{blue}) = 6/20 = 0.3$.

Definition 1.2 (Weighted k-NN, k-NN regression)**.** In **weighted k-NN** each neighbour votes with a weight inversely proportional to its distance:

$$p(y = c \mid \boldsymbol{x}, k) = \frac{1}{Z} \sum_{i \in \mathcal{N}_k(\boldsymbol{x})} \frac{\mathbb{I}(y_i = c)}{d(\boldsymbol{x}, \boldsymbol{x}_i)}, \qquad Z = \sum_{i \in \mathcal{N}_k(\boldsymbol{x})} \frac{1}{d(\boldsymbol{x}, \boldsymbol{x}_i)},$$

and again $\hat{y} = \arg\max_c \; p(y = c \mid \boldsymbol{x}, k)$. Dividing by Z makes the weights sum to 1:

$$w_i = \frac{1}{Z \, d(\boldsymbol{x}, \boldsymbol{x}_i)}, \qquad \sum_{i \in \mathcal{N}_k(\boldsymbol{x})} w_i = \frac{Z}{Z} = 1, \qquad p(y = c \mid \boldsymbol{x}, k) = \sum_{i \in \mathcal{N}_k(\boldsymbol{x}) \colon \, y_i = c} w_i,$$

so $p(y = c \mid \boldsymbol{x}, k)$ is the total weight of the neighbours of class c, and again a probability mass function. In **k-NN regression** ($y_i \in \mathbb{R}$) the prediction is the mean of the neighbours' values with the same weights:

$$\hat{y} = \frac{1}{Z} \sum_{i \in \mathcal{N}_k(\boldsymbol{x})} \frac{y_i}{d(\boldsymbol{x}, \boldsymbol{x}_i)} = \sum_{i \in \mathcal{N}_k(\boldsymbol{x})} w_i \, y_i.$$

Effect of k. With $k = 1$, an isolated training point lying among points of another class (e.g. a mislabelled one) gets its own small island in the decision regions. With $k = 3$ such a point is outvoted by its two other neighbours, so these islands mostly disappear and the decision boundaries become smoother (fig. 1). How to choose k is the topic of the next subsection.

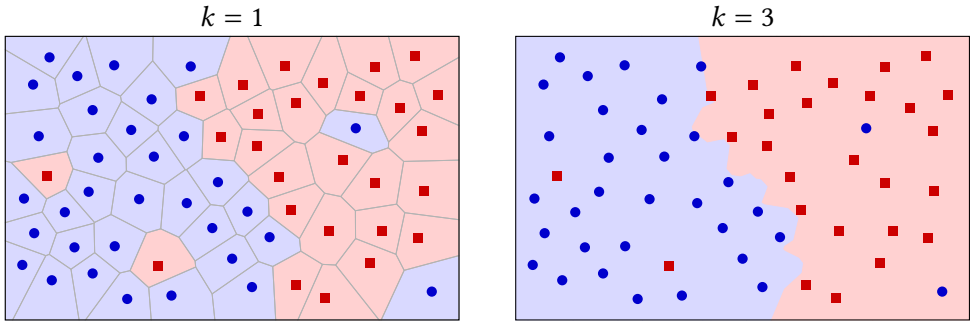


Figure 1: Decision regions of 1-NN (with the Voronoi cells) and 3-NN on the same two-class data.

1.2 Choosing k and measuring performance

Train, validation, test. The number of neighbours k is a **hyper-parameter**. The goal is **generalisation**: pick the k that performs best, in a predefined metric such as accuracy, on unseen future data. Future data are not available, so $\mathcal{D}$ is split (fig. 2) into a **learning set** $\mathcal{D}_L$ and a **test set** $\mathcal{D}_t$, and $\mathcal{D}_L$ further into a **training set** $\mathcal{D}_T$ and a **validation set** $\mathcal{D}_V$. **Hyper-parameter tuning** then goes as follows:

1. learn the model on $\mathcal{D}_T$;
2. evaluate it on $\mathcal{D}_V$ for each candidate k and keep the best one;
3. report the final performance on $\mathcal{D}_t$.

In practice one plots the validation accuracy against k and takes the maximiser: in one experiment it is about 0.9 for small k, maximal (about 0.93) at $k = 7$ and down to about 0.73 at $k = 34$, so $k = 7$ is chosen. Cross-validation and further good practice for model selection follow with decision trees (section 2.3).

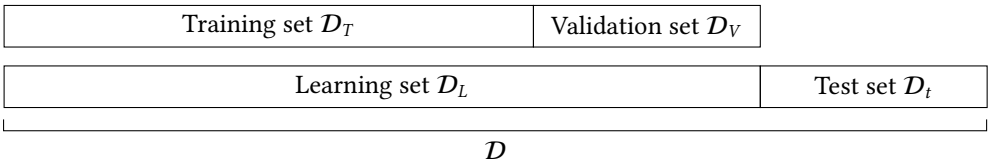
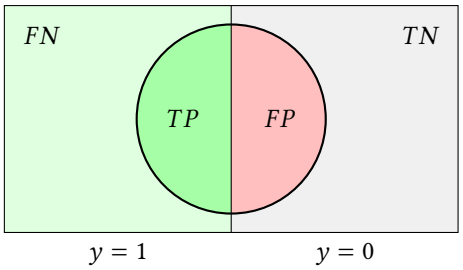


Figure 2: Splitting the dataset $\mathcal{D}$ for hyper-parameter tuning.

Binary classification metrics. For $y \in \{0, 1\}$, the predictions on a labelled set are counted in the **confusion table** (table 1 and fig. 3): **true positives** TP and **true negatives** TN are the correct predictions, **false positives** FP and **false negatives** FN the wrong ones.

	True condition	
Predicted	$y = 1$	$y = 0$
$\hat{y} = 1$	TP	FP
$\hat{y} = 0$	FN	TN

Table 1: Confusion table of a binary classifier.



disc: predicted $\hat{y} = 1$

$$\text{prec} = \frac{TP}{\text{disc}}, \quad \text{rec} = \frac{TP}{\text{left half}}$$

Figure 3: The four counts as regions: positives on the left, negatives on the right.

Definition 1.3 (Classification metrics)**.**

$$\text{accuracy:} \quad \text{acc} = \frac{TP + TN}{TP + TN + FP + FN}, \qquad \text{F1 score:} \quad f1 = \frac{2 \cdot \text{prec} \cdot \text{rec}}{\text{prec} + \text{rec}},$$

$$\text{precision:} \quad \text{prec} = \frac{TP}{TP + FP}, \qquad \text{sensitivity/recall:} \quad \text{rec} = \frac{TP}{TP + FN},$$

$$\text{specificity:} \quad \text{tnr} = \frac{TN}{FP + TN}, \qquad \text{false negative rate:} \quad \text{fnr} = \frac{FN}{TP + FN},$$

$$\text{false positive rate:} \quad \text{fpr} = \frac{FP}{FP + TN}.$$

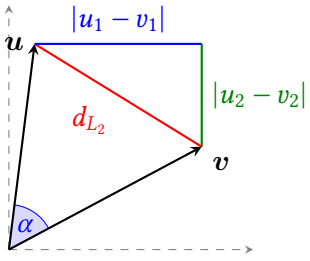
The F1 score is the harmonic mean of precision and recall, $f1 = 2\Big/\left(\frac{1}{\text{prec}} + \frac{1}{\text{rec}}\right)$.

Precision and recall trade off: increasing one most often decreases the other. All these numbers must be read with care when the classes are imbalanced.

1.3 Distances, scaling and dimensionality

Distance measures. k-NN works with any distance measure, which makes it highly flexible. For $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^D$ (fig. 4):

- **Euclidean** (L_2) distance $\|\boldsymbol{u} - \boldsymbol{v}\|_2 = \sqrt{\sum_j (u_j - v_j)^2}$;
- L_1 distance $\sum_j |u_j - v_j|$ and L_∞ distance $\max_j |u_j - v_j|$;
- the **angle** α between them, $\cos\alpha = \dfrac{\boldsymbol{u}^T \boldsymbol{v}}{\|\boldsymbol{u}\|\|\boldsymbol{v}\|}$;
- the **Mahalanobis distance** $\sqrt{(\boldsymbol{u} - \boldsymbol{v})^T \boldsymbol{\Sigma}^{-1} (\boldsymbol{u} - \boldsymbol{v})}$, with $\boldsymbol{\Sigma}$ symmetric positive definite;
- Hamming distance, edit distance, and others.



$$d_{L_1} = |u_1 - v_1| + |u_2 - v_2|$$
$$d_{L_2} = \sqrt{(u_1 - v_1)^2 + (u_2 - v_2)^2}$$
$$d_{L_\infty} = \max_j |u_j - v_j| = |u_1 - v_1|$$

Figure 4: L_1, L_2, L_∞ distances and angle α between two vectors of $\mathbb{R}^2$.

Scaling. A distance mixes all features, so features on very different scales distort it. For instance, if in $\mathbb{R}^2$ the feature x_1 is of the order of metres and x_2 of the order of centimetres (say x_1 up to 1000 cm, x_2 below 6 cm), the distance, hence the nearest neighbour, is determined almost only by x_1: the 1-NN decision regions become stripes orthogonal to the x_1-axis and x_2 is effectively ignored. Two remedies:

- **standardisation**: scale each feature j to zero mean and unit variance, $x_{j,\text{std}} = \dfrac{x_j - \mu_j}{\sigma_j}$, with μ_j, σ_j the mean and standard deviation of feature j; this is standard practice in machine learning, since many models are sensitive to differences in scale;
- the Mahalanobis distance with $\boldsymbol{\Sigma} = \text{diag}(\sigma_1^2, \dots, \sigma_D^2)$, which equals the Euclidean distance between the standardised data.

Indeed $\boldsymbol{\Sigma}^{-1} = \text{diag}(\sigma_1^{-2}, \dots, \sigma_D^{-2})$ and the means cancel in the differences, so

$$(\boldsymbol{u} - \boldsymbol{v})^T \boldsymbol{\Sigma}^{-1} (\boldsymbol{u} - \boldsymbol{v}) = \sum_{j=1}^{D} \frac{(u_j - v_j)^2}{\sigma_j^2} = \sum_{j=1}^{D} \left(\frac{u_j - \mu_j}{\sigma_j} - \frac{v_j - \mu_j}{\sigma_j} \right)^2 = \|\boldsymbol{u}_{\text{std}} - \boldsymbol{v}_{\text{std}}\|_2^2.$$

Curse of dimensionality. Draw $N = 20$ samples uniformly from the discrete input space $\{1, \dots, 10\}^D$, which has 10^D points; the samples occupy at most 20 of them. For $D = 1$ they cover most or all of the 10 points, for $D = 2$ at most 20% of the 100 points (about 18%, since some samples coincide), and for $D = 3$ at most 2% of the 1000 points. Hence in high dimension the nearest neighbour of a new point is typically far away, and to cover the same fraction of the input space N has to grow exponentially with the number of features: this is the **curse of dimensionality**, to keep in mind when using k-NN on high-dimensional data.

Cost. Naive k-NN is expensive: memory and inference are both $O(N)$ (for fixed D), since the whole training set is stored and every query is compared with all training points. Tree-based search structures such as the **k-d tree** give efficient (approximate) nearest-neighbour search, at the price of an additional computation performed only once (building the structure).

2 Decision Trees

2.1 Decision trees and inference

Structure. A decision tree classifies by asking a sequence of questions, as in the game of twenty questions: "does it have fur?", if yes "does it bark?" (dog or cat), if no "does it fly?" (bird or fish). In a **decision tree** each inner **node** is a test on a feature, each **branch** leaving it is one outcome of the test, and each **leaf** is a region of the input space together with the distribution of the training labels in that region. Figure 5 shows a tree for the tennis dataset, whose goal is to predict *PlayTennis* for unseen days: all four overcast days are "Yes", so that branch is a leaf right away, while sunny days need a test on Humidity and rainy days one on Windy. The tree classifies all 14 training days correctly.

Outlook	Temperature	Humidity	Windy	PlayTennis
Sunny	Hot	High	False	No
Sunny	Hot	High	True	No
Overcast	Hot	High	False	Yes
Rainy	Mild	High	False	Yes
Rainy	Cool	Normal	False	Yes
Rainy	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Sunny	Mild	High	False	No
Sunny	Cool	Normal	False	Yes
Rainy	Mild	Normal	False	Yes
Sunny	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Rainy	Mild	High	True	No

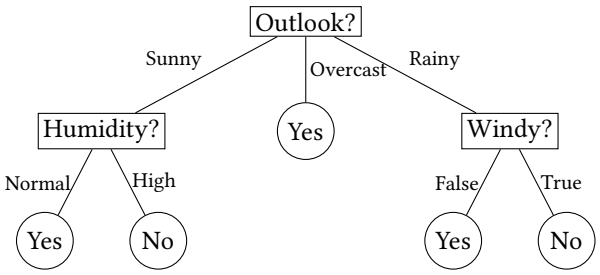


Figure 5: Tennis dataset and a decision tree that classifies it without errors.

Numerical features. For $\boldsymbol{x} \in \mathbb{R}^D$ the simplest test is a **binary split** on a single feature, $x_j \leq a$ (true) or $x_j > a$ (false); in this lecture these are the only tests on numerical features. Each split cuts a region with an axis-parallel hyperplane, so a decision tree partitions the input space into cuboid (axis-aligned box) regions, one per leaf (fig. 6).

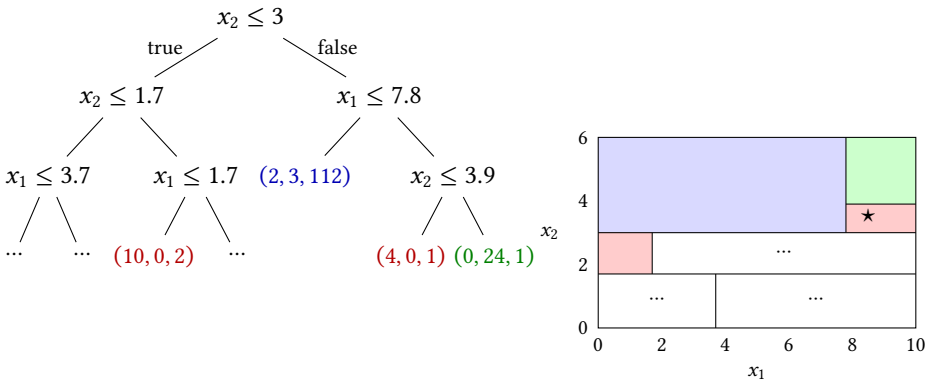


Figure 6: Top of a tree on three classes (leaf counts (n_1, n_2, n_3), unexpanded subtrees as $\cdots$) and its partition of $[0, 10] \times [0, 6]$; the star is $\boldsymbol{x} = (8.5, 3.5)^T$.

Definition 2.1 (Inference in a decision tree)**.** Let the classes be $\{1, \dots, C\}$. To classify $\boldsymbol{x}$, test its features from the root down to the leaf whose region $\mathcal{R}$ contains $\boldsymbol{x}$, and let $\boldsymbol{n}_\mathcal{R} = (n_{1,\mathcal{R}}, \dots, n_{C,\mathcal{R}})$ be the class counts of the training points in $\mathcal{R}$. Then

$$p(y = c \mid \mathcal{R}) = \frac{n_{c,\mathcal{R}}}{\sum_{c'=1}^{C} n_{c',\mathcal{R}}}, \qquad \hat{y} = \arg\max_c \; p(y = c \mid \boldsymbol{x}) = \arg\max_c \; p(y = c \mid \mathcal{R}) = \arg\max_c \; n_{c,\mathcal{R}},$$

i.e. $\boldsymbol{x}$ gets the majority label of its region, as in k-NN (definition 1.1).

Example 2.2. In fig. 6, $\boldsymbol{x} = (8.5, 3.5)^T$ has $x_2 > 3$, then $x_1 > 7.8$, then $x_2 \leq 3.9$, and reaches the leaf with counts $(4, 0, 1)$. Hence $p(y \mid \boldsymbol{x}) = (\frac{4}{5}, 0, \frac{1}{5})$ and $\hat{y} = 1$.

2.2 Growing a tree: splitting criteria

The optimal tree is intractable. As for k-NN, the goal is generalisation, and the tree is chosen with the split of fig. 2: build trees on $\mathcal{D}_T$, predict the labels of $\mathcal{D}_V$, keep the tree with the best validation performance and report its performance on $\mathcal{D}_t$. The naive idea is to build all possible trees. Already the tests are many: a feature with n distinct values in the training set has $n - 1$ thresholds a that split the data differently (one between each pair of consecutive values), so 2 features with 300 distinct values each give $2 \cdot 299 = 598$ possible tests at the root, slightly fewer at each descendant. The number of trees explodes, and finding the optimal tree is NP-complete (optimal meaning that it minimises the expected number of tests needed to classify an unknown sample; even identifying the root node of an optimal tree is NP-hard). Instead, the tree is grown top-down on the training data, choosing the best split node by node with a **greedy heuristic**.

Definition 2.3 (Impurity improvement)**.** Let $i(t)$ be a splitting criterion measuring the impurity of the class distribution $\pi_c = p(y = c \mid t)$ of the training points at node t. A split s sends a fraction p_L of them to the left child t_L and $p_R = 1 - p_L$ to the right child t_R. The **improvement** of s is

$$\Delta i(s, t) = i(t) - p_L \, i(t_L) - p_R \, i(t_R).$$

At each node the tree is split with the test (over all features and thresholds) of largest $\Delta i(s, t)$.

Misclassification rate. The first candidate is the **misclassification rate** $i_E(t) = 1 - \max_c \pi_c$, the training error of the node when it predicts its majority class. For instance, a node with 150 red, 100 blue and 50 green points has $i_E(t) = 1 - \frac{150}{300} = 0.5$; the split $x_1 \le 5$ of fig. 7(a) leaves a pure left child and a right child with $i_E(t_R) = 1 - \frac{100}{150} = \frac{1}{3}$, so $\Delta i_E = 0.5 - \frac{1}{2} \cdot 0 - \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{3}$. The misclassification rate has two problems:

1. *no split may be performed although two splits together would classify perfectly.* In fig. 7(b), 40 of 200 points are red, so $i_E(t) = \frac{40}{200}$. The split $x_1 \le 5$ gives a pure child and a child with 40 red out of 100: $p_L\, i_E(t_L) + p_R\, i_E(t_R) = \frac{1}{2} \cdot 0 + \frac{1}{2} \cdot \frac{40}{100} = \frac{40}{200}$, and the same holds for $x_2 \le 3$. Hence $\Delta i_E = 0$ for both splits and none is made, although $x_1 > 5$ and $x_2 \le 3$ together isolate the red class;
2. *it is insensitive to changes in the class probabilities.* Starting from class counts $(400, 400)$, so $i_E(t) = \frac{1}{2}$, the split a into $(100, 300)$ and $(300, 100)$ and the split b into $(200, 400)$ and $(200, 0)$ have the same improvement,

$$\Delta i_E(a, t) = \tfrac{1}{2} - \tfrac{1}{2} \cdot \tfrac{1}{4} - \tfrac{1}{2} \cdot \tfrac{1}{4} = 0.25, \qquad \Delta i_E(b, t) = \tfrac{1}{2} - \tfrac{3}{4} \cdot \tfrac{1}{3} - \tfrac{1}{4} \cdot 0 = 0.25,$$

   although b produces a pure node.

On the data of fig. 6 the same flaw shows up: as a function of the threshold, Δi_E is exactly 0 for most thresholds of x_1 and x_2, and positive (at most about 0.04) only for x_1 between about 6.5 and 10 and for x_2 close to 3.

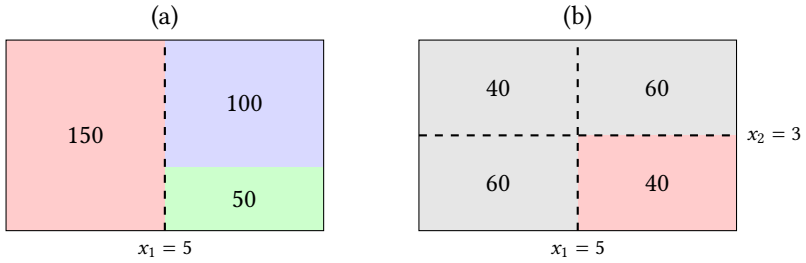


Figure 7: Class counts of two nodes: (a) a useful split $x_1 \le 5$; (b) data where neither $x_1 \le 5$ nor $x_2 \le 3$ improves i_E.

Impurity measures. A suitable criterion $i(t)$ should be *maximal* when the classes are equally distributed in the node, *minimal* (usually 0) when the node is pure, and *symmetric* in the classes.

Definition 2.4 (Impurity measures)**.** With $\pi_c = p(y = c \mid t)$:

$$i_E(t) = 1 - \max_c \pi_c, \qquad i_H(t) = -\sum_{c=1}^{C} \pi_c \log_2 \pi_c, \qquad i_G(t) = \sum_{c=1}^{C} \pi_c(1 - \pi_c) = 1 - \sum_{c=1}^{C} \pi_c^2,$$

the **misclassification rate**, the **entropy** (with $0 \log_2 0 = 0$, the limit of $x \log x$ as $x \to 0^+$) and the **Gini index**. The two forms of i_G agree because $\sum_c \pi_c(1 - \pi_c) = \sum_c \pi_c - \sum_c \pi_c^2$ and $\sum_c \pi_c = 1$.

For two classes, with $\pi = \pi_1$ and $\pi_2 = 1 - \pi$ (fig. 8),

$$i_E = 1 - \max(\pi, 1 - \pi) = \min(\pi, 1 - \pi), \qquad i_G = 1 - \pi^2 - (1 - \pi)^2 = 2\pi(1 - \pi),$$
$$i_H = -\pi \log_2 \pi - (1 - \pi) \log_2 (1 - \pi) \ :$$

all three vanish at $\pi \in \{0, 1\}$ and are maximal at $\pi = \frac{1}{2}$, with maximum $\frac{1}{2}$ for i_E, i_G and 1 for i_H.

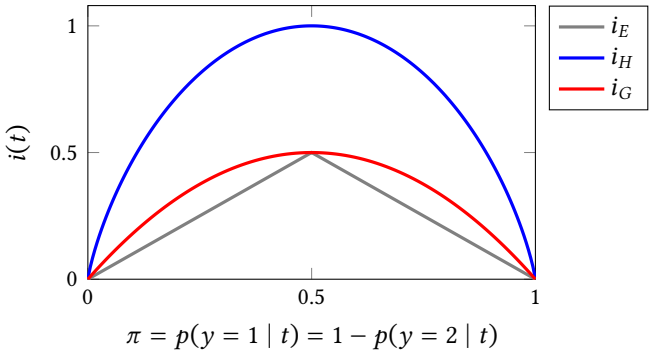


Figure 8: Misclassification rate, entropy and Gini index for two classes.

Remark. Entropy and Gini index are strictly concave in π, while i_E is linear as long as the majority class does not change; this is why they fix both problems. With i_G, in (b) of fig. 7 either split has $\Delta i_G > 0$, and b is preferred to a:

$$i_G(t) = 1 - 0.8^2 - 0.2^2 = 0.32, \qquad \Delta i_G = 0.32 - \tfrac{1}{2} \cdot 0 - \tfrac{1}{2}\,(1 - 0.6^2 - 0.4^2) = 0.08,$$
$$\Delta i_G(a, t) = \tfrac{1}{2} - \tfrac{3}{8} = 0.125 \;<\; \Delta i_G(b, t) = \tfrac{1}{2} - \tfrac{3}{4} \cdot \tfrac{4}{9} = \tfrac{1}{6}.$$

Entropy and information. For a discrete random variable X with values $\{x_1, \dots, x_n\}$ the **Shannon entropy** is

$$\mathbb{H}(X) = -\sum_{i=1}^{n} p(X = x_i) \log_2 p(X = x_i),$$

the expected number of bits needed to encode a randomly drawn value of X under the most efficient code. For a variable with 4 values it is 2 bits for the uniform distribution ($-4 \cdot \frac{1}{4} \log_2 \frac{1}{4}$), 0 when one value has probability 1, and in between (1.9333 in one non-uniform case) otherwise: higher entropy means a flatter histogram and less predictable values, lower entropy a peakier histogram and more predictable values. As an example of efficient coding, take four messages with probabilities 0.01 ("there is free beer"), 0.02 ("you have an exam"), 0.30 ("you have a lecture") and 0.67 ("nothing happening"). The **Huffman code** gives shorter codewords to likelier messages, here 111, 110, 10, 0, with average length

$$0.01 \cdot 3 + 0.02 \cdot 3 + 0.30 \cdot 2 + 0.67 \cdot 1 = 1.36 \text{ bits}.$$

Remark. In general $\mathbb{H}(X)$ is a lower bound on the average code length, reached when all probabilities are powers of $\frac{1}{2}$; here $\mathbb{H} \approx 1.09 < 1.36$.

Gini index. $i_G(t) = \sum_c \pi_c(1 - \pi_c)$ is the probability that a randomly chosen instance of the node is misclassified when it is labelled at random according to the node's class distribution: π_c is the probability of picking an instance of class c, and $1 - \pi_c$ the probability that its random label is wrong. In practice the choice between entropy and Gini index matters in only about 2% of the cases (Raileanu and Stoffel); the Gini index has the small advantage of needing no logarithm, so it can be a bit faster to compute.

Example 2.5. The data of fig. 6 have 300 points with class counts $(67, 65, 168)$, so at the root

$$i_G(t) = 1 - \left(\frac{67}{300}\right)^2 - \left(\frac{65}{300}\right)^2 - \left(\frac{168}{300}\right)^2 \approx 0.5896.$$

The test $x_2 \le 3$ sends 153 points to t_L, with counts $(61, 38, 54)$ and $i_G(t_L) \approx 0.6548$, and 147 to t_R, with counts $(6, 27, 114)$ and $i_G(t_R) \approx 0.3632$. Hence

$$\Delta i_G(x_2 \le 3, t) = i_G(t) - \frac{153}{300}\, i_G(t_L) - \frac{147}{300}\, i_G(t_R) \approx 0.07768,$$

the largest improvement over all tests, so $x_2 \le 3$ becomes the root. As a function of the threshold, Δi_G is positive and varies smoothly for all thresholds of both features (up to about 0.035 for x_1), unlike Δi_E.

Growing the tree. Applying the greedy step recursively, the label distributions of the nodes become progressively purer (mostly instances of one class). In fig. 6 only the root split $x_2 \leq 3$ is the greedy choice (example 2.5); the lower splits just illustrate the partition. On the training data the accuracy of the greedy Gini tree grows with its depth: 58.3% at depth 1, 77% at depth 2, 84.3% at depth 3, 90.3% at depth 4, and 100% for the **maximally pure** tree, whose leaves are all pure and whose decision regions include many small boxes around single points. Perfect training accuracy does not mean good generalisation.

2.3 Overfitting, pruning and extensions

Overfitting. **Overfitting** typically occurs when the model tries to fit the training data perfectly, and means poor generalisation. It is recognised by a low training error (possibly 0) together with a comparably high validation error. For decision trees, training accuracy never decreases with a split: since the class counts of t are the sums of those of its children, $n_{c,t} = n_{c,t_L} + n_{c,t_R}$, the number of correctly classified training points can only grow,

$$\max_c n_{c,t} \leq \max_c n_{c,t_L} + \max_c n_{c,t_R}.$$

The validation accuracy tells how well the model generalises. In a typical experiment, as the number of leaves grows from 2 to about 35, the training accuracy rises steadily from about 0.6 to 1, while the validation accuracy rises with it at first (**underfitting** with very few leaves), peaks at about 0.88 with 16 leaves and then drops to about 0.77 (overfitting). In practice, with increasing model size, data size or training time, the performance sometimes first improves, then gets worse, and then improves again: this **double descent phenomenon** was observed recently and is actively studied.

Model selection. Model selection and the fight against overfitting use the split of fig. 2: models are built on the training set and compared on the validation set, and the test set is touched only once, at the end, to report the final performance.

Definition 2.6 (K-fold cross-validation)**.** Split the learning set $\mathcal{D}_L$ into K disjoint folds $\mathcal{D}_{V_1}, \ldots, \mathcal{D}_{V_K}$ ($K = 10$ is common) and let $\mathcal{D}_{T_k} = \mathcal{D}_L \setminus \mathcal{D}_{V_k}$ (fig. 9). For each k, train on the $K-1$ folds $\mathcal{D}_{T_k}$ and evaluate on $\mathcal{D}_{V_k}$; the average over the folds,

$$\mathrm{err}_{\mathrm{CV}} = \frac{1}{K} \sum_{k=1}^{K} \mathrm{err}_{\mathcal{D}_{V_k}}(f_k), \qquad f_k \text{ trained on } \mathcal{D}_{T_k},$$

estimates the error of a setting of the hyper-parameters (or of the chosen model, for model selection). Repeat for different settings of the hyper-parameters, then train with the best setting on all of $\mathcal{D}_L$ and test on $\mathcal{D}_t$.

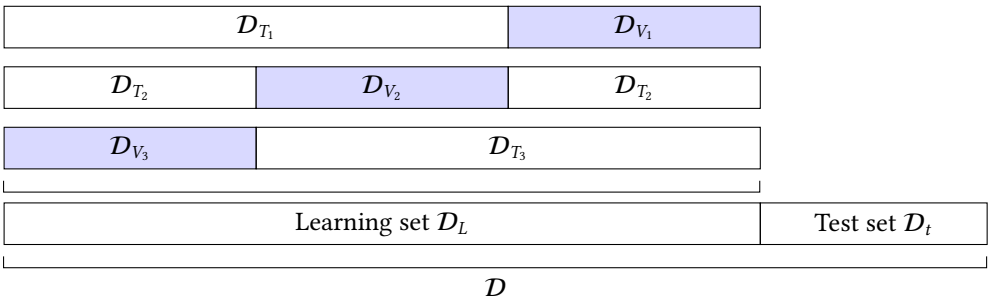


Figure 9: 3-fold cross-validation on the learning set; validation folds shaded.

In **leave-one-out cross-validation** (LOOCV) each model is trained on all but one sample: with N samples it is N-fold cross-validation. It is useful when data are scarce, since it uses as much data as

possible for training and still gives a good estimate of the performance, but the model has to be trained N times. With large amounts of data and an expensive model, a smaller K or a single validation set is the better choice.

Stopping criteria. The tree is grown by recursive splitting, which must stop at some point. Possible **stopping** (or **pre-pruning**) criteria: the distribution in the branch is pure, $i(t) = 0$; a maximum depth is reached; the number of samples in each branch falls below a threshold t_n; the benefit of splitting falls below a threshold, $\Delta i(s, t) < t_\Delta$; the accuracy on the validation set. Alternatively, grow the tree maximally and then (post-)prune it.

Definition 2.7 (Reduced error pruning)**.** Let T be a decision tree, t an inner node and T_t the subtree rooted at t. **Pruning** T w.r.t. t deletes all descendants of t (but not t, which becomes a leaf); the pruned tree is written $T \setminus T_t$ (fig. 10). **Reduced error pruning** uses the error $\mathrm{err}_{D_V}(T)$ on the validation set:

1. for each inner node t compute $\mathrm{err}_{D_V}(T \setminus T_t)$;
2. prune T at the node that yields the largest error reduction;
3. repeat until $\mathrm{err}_{D_V}(T) < \mathrm{err}_{D_V}(T \setminus T_t)$ for all inner nodes t.

After pruning, both training and validation data may be used to update the labels at each leaf.

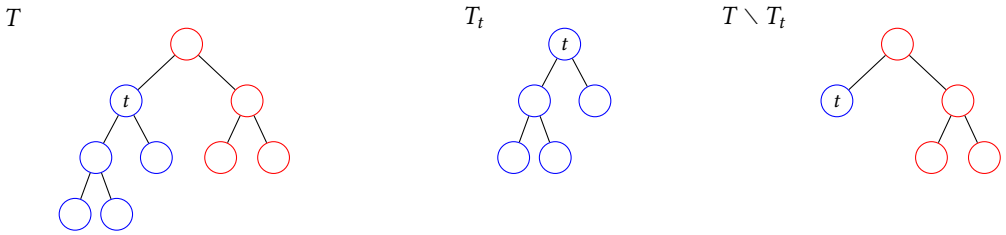


Figure 10: A tree T, its subtree T_t and the pruned tree $T \setminus T_t$.

Categorical features and regression. Categorical features can be tested by equality, e.g. "Outlook = overcast?" (yes: play), otherwise "Temperature = hot?" (yes: do not play), otherwise "Wind = weak?", and so on, or by one branch per value as in fig. 5; the algorithm variants ID3, C4.5 and CART handle them differently. For **regression** ($y_i \in \mathbb{R}$), a leaf predicts the mean of the training outputs in its region instead of the mode, and the splitting heuristic is the mean squared error, i.e. definition 2.3 with $i(t) = \frac{1}{n_t} \sum_{i \in t} (y_i - \bar{y}_t)^2$, where n_t is the number of training points at t and $\bar{y}_t$ their mean output. The prediction is then a piecewise constant function of $\boldsymbol{x}$; on one-dimensional data a tree of depth 5 follows the points much more closely than one of depth 2.

Properties. Decision trees are human interpretable and handle any combination of numerical and categorical features and targets; their extensions (random forests, boosted trees) have very competitive performance, e.g. in Kaggle competitions. Compared to k-NN (section 1) they have a much better complexity in memory/storage and inference, since only the tree is stored and a query follows a single root-to-leaf path, and a more flexible decision function.

Ensembles. An **ensemble** aggregates the predictions of many (diverse) classifiers to improve the performance; the main benefit is a lower variance of the model, obtained by averaging (see the bias-variance trade-off in the lecture on linear regression). Main types:

- **bagging** (bootstrap aggregating): create new datasets by sampling the training set with replacement, train a separate classifier on each and combine their predictions, e.g. by average or majority vote;
- **boosting**: train (weak) classifiers incrementally, each one correcting the previous mistakes by giving higher weight to hard (misclassified) examples;

- **stacking**: train a meta-classifier that takes the base classifiers' predictions as features;
- further approaches: bucket of models, Bayesian model averaging, Bayes optimal classifier.

Random forests. Bagged decision trees can be highly correlated, which reduces the benefit of bagging since it needs diverse classifiers. A **random forest** therefore learns each tree on a random subset of the features as well, i.e. bagging at instance level plus bagging at feature level. Random forests are widely used because of their good "out-of-the-box" performance (not much tuning required), but are less interpretable than a single tree. Rule of thumb for the number of random features: $\log_2 D$ for regression and $\sqrt{D}$ for classification, with D the number of features.

Boosting: AdaBoost and XGBoost. Boosting trains "weak learners" incrementally, e.g. **decision stumps** (one-level trees):

1. initialise a weight vector over the training examples with uniform weights;
2. loop: train a weak learner on the weighted examples (e.g. by weighted sampling), then increase the weights of the misclassified examples;
3. predict by the (error-based) weighted majority of the weak learners.

Adaptive boosting (**AdaBoost**) and gradient boosting (**XGBoost**) differ mainly in how the weights are updated: AdaBoost uses closed-form weight updates based on the errors, XGBoost adds one tree, and one level in the tree, at a time, splitting greedily based on the gradient w.r.t. a (custom) loss function.

3 Probabilistic Inference

3.1 Maximum likelihood estimation

The coin problem. A coin is flipped 10 times and shows H T H H T H H H T H (3 tails, 7 heads). What is the probability that the next flip is T? The answer 30% seems reasonable; the aim is to justify it. Every flip is random, so every sequence of flips has some probability of being observed. Flip i is a random variable $F_i \in \{\mathrm{T}, \mathrm{H}\}$ with

$$p_i(F_i = \mathrm{T} \mid \theta_i) = \mathrm{Ber}(F_i = \mathrm{T} \mid \theta_i) = \theta_i, \qquad \text{i.e. } F_i \sim \mathrm{Ber}(\theta_i),$$

where the notation $\mid \theta_i$ says that the distribution depends on the parameter θ_i. The question is about θ_{11}, while the observed sequence is governed by $\theta_1, \dots, \theta_{10}$: without further assumptions the data say nothing about θ_{11}.

Likelihood. A natural principle is to choose the parameters under which the observation is as probable as possible, i.e. to **maximise the likelihood** of the observation. This requires a model of $p(\mathcal{D} \mid \theta_1, \dots, \theta_{10})$, built from two assumptions:

- *identical distribution*: the flips are qualitatively the same, $p_i(F_i = f_i \mid \theta_i) = p(F_i = f_i \mid \theta)$ with θ independent of i;
- *independence*: the flips do not affect each other, so the probability of the sequence is the product of the probabilities of the single flips.

Together, the flips are **independent and identically distributed** (**i.i.d.**). This links the unknown θ_{11} to the observed flips, since $\theta_{11} = \theta$ as well.

Definition 3.1 (Likelihood, maximum likelihood estimation)**.** Let $\mathcal{D} = (f_1, \dots, f_N)$ be i.i.d. flips with $p(F = \mathrm{T} \mid \theta) = \theta$ and $p(F = \mathrm{H} \mid \theta) = 1 - \theta$, i.e. $p(F = f \mid \theta) = \theta^{\mathbb{I}(f=\mathrm{T})}(1-\theta)^{\mathbb{I}(f=\mathrm{H})}$, and let $|T|, |H|$ be the numbers of tails and heads in $\mathcal{D}$. Then

$$p(\mathcal{D} \mid \theta) = \prod_{i=1}^{N} p(F_i = f_i \mid \theta) = \theta^{|T|}(1-\theta)^{|H|}.$$

As a function of θ, $f(\theta) := p(\mathcal{D} \mid \theta)$ is the **likelihood function**. The **maximum likelihood estimate** (**MLE**) is

$$\theta_{\text{MLE}} = \arg\max_{\theta \in [0,1]} p(\mathcal{D} \mid \theta).$$

For the sequence above, $p(\mathcal{D} \mid \theta) = (1-\theta)\,\theta\,(1-\theta)(1-\theta)\,\theta\,(1-\theta)(1-\theta)(1-\theta)\,\theta\,(1-\theta) = \theta^3(1-\theta)^7$. The likelihood is *not* a probability distribution over θ: in general $\int p(\mathcal{D} \mid \theta)\,\mathrm{d}\theta \neq 1$, and here $\int_0^1 \theta^3(1-\theta)^7\,\mathrm{d}\theta = \frac{3!\,7!}{11!} = \frac{1}{1320}$ (by the normalisation of the Beta density, definition 3.4).

Maximising the likelihood. One can set $\frac{\mathrm{d}f}{\mathrm{d}\theta} = 0$, solve for θ and check the critical points with the second derivative, but even for this simple f the algebra is ugly. It simplifies with the logarithm: log is strictly increasing, so f and $\log f$ have the same maximisers (and the same critical points, since $(\log f)' = f'/f$ where $f > 0$),

$$\arg\max_{\theta \in [0,1]} f(\theta) = \arg\max_{\theta \in [0,1]} \log f(\theta),$$

and the logarithm turns the product into a sum (fig. 11).

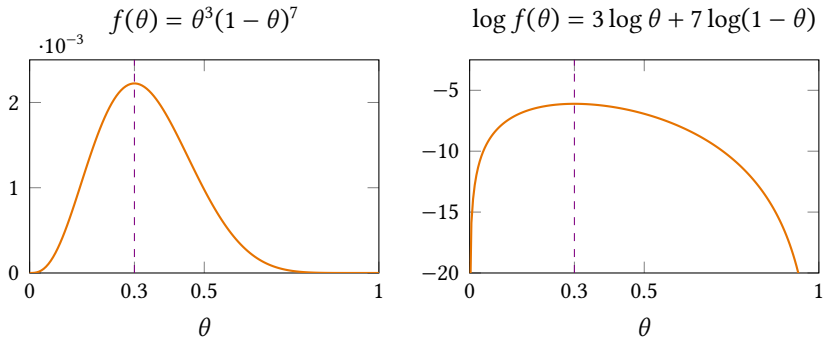


Figure 11: Likelihood and log-likelihood of the coin sequence.

For any sequence, $\log p(\mathcal{D} \mid \theta) = |T| \log\theta + |H| \log(1-\theta)$, and

$$\frac{\mathrm{d}}{\mathrm{d}\theta} \log p(\mathcal{D} \mid \theta) = \frac{|T|}{\theta} - \frac{|H|}{1-\theta} = 0 \iff |T|(1-\theta) = |H|\,\theta \iff \theta = \frac{|T|}{|T| + |H|},$$

a maximum because the second derivative $-\frac{|T|}{\theta^2} - \frac{|H|}{(1-\theta)^2}$ is negative. Hence

$$\theta_{\text{MLE}} = \frac{|T|}{|T| + |H|}.$$

The next flip is then modelled as $F_{11} \sim \text{Ber}(\theta_{\text{MLE}})$, so $p(F_{11} = \text{T} \mid \theta_{\text{MLE}}) = \theta_{\text{MLE}} = \frac{3}{10}$: this justifies the answer 30%.

Limits of the MLE. Suppose the same coin is flipped twice and shows H H. Then $p(\mathcal{D} \mid \theta) = (1-\theta)^2$ is maximal at $\theta_{\text{MLE}} = 0$: the MLE declares tails impossible, although even a fair coin ($\theta = 0.5$) shows H H with probability $0.5^2 = 25\%$. The result is counter-intuitive because we hold *prior beliefs* (coins usually do not land heads all the time), which the MLE ignores; for this reason it performs poorly when little data is available. The beliefs have to be represented mathematically and incorporated into the model.

3.2 Bayesian inference and MAP estimation

Prior. In the (subjective) Bayesian interpretation of probability, a distribution $p(\theta)$ reflects our subjective beliefs about θ. A **prior distribution** $p(\theta)$ represents the beliefs before any data are observed.

Its only constraints are that it must not depend on the data, that $p(\theta) \geq 0$ and that $\int p(\theta)\,\mathrm{d}\theta = 1$, the last two on the support, i.e. the feasible values of θ (here $\theta \in [0,1]$). This leaves room for (possibly subjective) modelling choices: a uniform density on $[0,1]$, a bump centred at $\theta = 0.5$, a bump around a smaller value, and so on.

Definition 3.2 (Bayes' formula)**.** After observing the data $\mathcal{D}$, beliefs about θ are updated by

$$p(\theta \mid \mathcal{D}) = \frac{p(\mathcal{D} \mid \theta)\, p(\theta)}{p(\mathcal{D})}, \qquad p(\mathcal{D}) = \int p(\mathcal{D}, \theta)\,\mathrm{d}\theta = \int p(\mathcal{D} \mid \theta)\, p(\theta)\,\mathrm{d}\theta,$$

where $p(\theta \mid \mathcal{D})$ is the **posterior** distribution (the beliefs after observing the data), $p(\mathcal{D} \mid \theta)$ the **likelihood**, $p(\theta)$ the **prior** and $p(\mathcal{D})$ the **evidence**, obtained by the sum rule of probability. The evidence does not depend on θ and only normalises the posterior so that it integrates to 1:

$$\text{posterior} \propto \text{likelihood} \cdot \text{prior}, \qquad p(\theta \mid \mathcal{D}) \propto p(\mathcal{D} \mid \theta)\, p(\theta).$$

Usually a model is specified by the likelihood and the prior; the evidence follows from them. Since the posterior is proportional to the prior, a value with $p(\theta) = 0$ keeps posterior 0 whatever the likelihood, i.e. whatever data are observed.

MAP estimation. The MLE asks the wrong question: it maximises $p(\mathcal{D} \mid \theta)$ and ignores the prior beliefs. What matters is the posterior $p(\theta \mid \mathcal{D})$.

Definition 3.3 (Maximum a posteriori estimation)**.** The **maximum a posteriori** (**MAP**) estimate maximises the posterior:

$$\theta_{\mathrm{MAP}} = \arg\max_{\theta}\ p(\theta \mid \mathcal{D}) = \arg\max_{\theta}\ \frac{p(\mathcal{D} \mid \theta)\, p(\theta)}{p(\mathcal{D})} = \arg\max_{\theta}\ p(\mathcal{D} \mid \theta)\, p(\theta),$$

where the factor $\frac{1}{p(\mathcal{D})}$ can be dropped because it is a positive constant independent of θ.

The likelihood is known; the prior is often chosen to make the subsequent calculations easier. For the coin the choice is the Beta distribution, for reasons that become clear when the full posterior is computed (section 3.3).

Definition 3.4 (Beta distribution)**.** For parameters $a > 0$, $b > 0$,

$$\mathrm{Beta}(\theta \mid a, b) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}\, \theta^{a-1}(1-\theta)^{b-1}, \qquad \theta \in [0,1],$$

where Γ is the **gamma function**, with $\Gamma(n) = (n-1)!$ for $n \in \mathbb{N}$. The constant makes the density integrate to 1, i.e. $\int_0^1 \theta^{a-1}(1-\theta)^{b-1}\,\mathrm{d}\theta = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$.

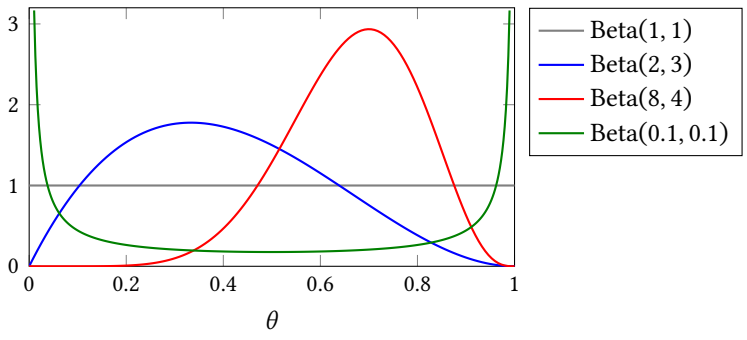


Figure 12: Beta densities.

Posterior and MAP for the coin. With $p(\mathcal{D} \mid \theta) = \theta^{|T|}(1-\theta)^{|H|}$ and the prior $p(\theta) = \mathrm{Beta}(\theta \mid a, b)$, Bayes' formula gives, dropping the constants independent of θ ($p(\mathcal{D})$ and $\frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}$),

$$p(\theta \mid \mathcal{D}) \propto \theta^{|T|}(1-\theta)^{|H|} \cdot \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}\, \theta^{a-1}(1-\theta)^{b-1} \propto \theta^{|T|+a-1}(1-\theta)^{|H|+b-1}. \tag{1}$$

Observing more data increases our confidence: with the prior Beta(6, 3), whose mode is at $\theta = \frac{5}{7}$, and data with $|T| = |H|/2$, the posterior lies between prior and likelihood and, as $|H|$ grows from 4 to 32, becomes narrower and moves towards the maximiser $\frac{1}{3}$ of the likelihood (fig. 13).

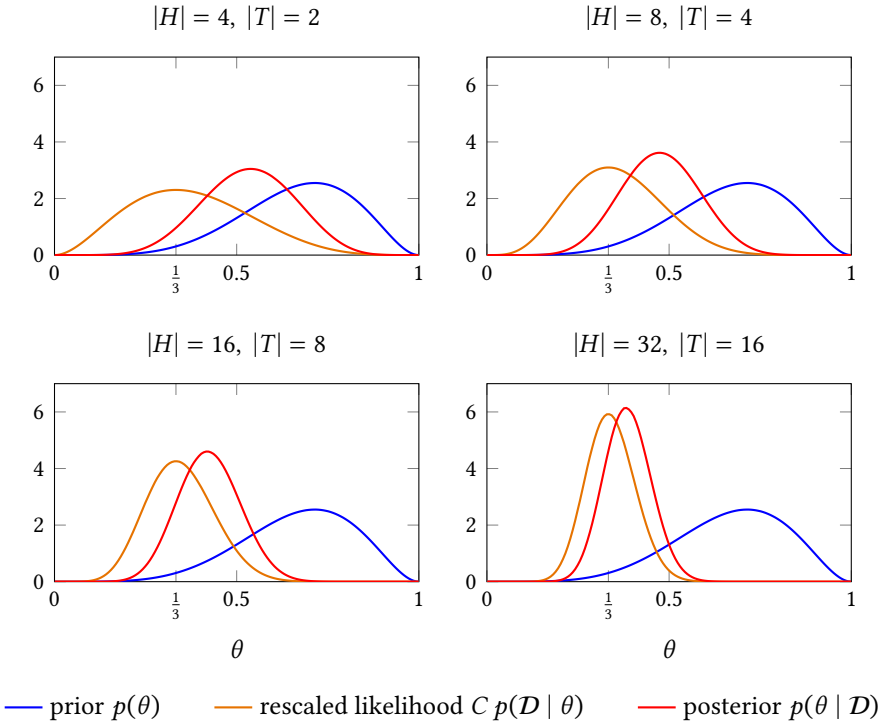


Figure 13: Prior, rescaled likelihood and posterior for growing data.

As for the MLE, take the logarithm:

$$\theta_{\mathrm{MAP}} = \arg\max_{\theta} \log p(\theta \mid \mathcal{D}) = \arg\max_{\theta} \big[(|T| + a - 1)\log\theta + (|H| + b - 1)\log(1-\theta)\big].$$

This is the log-likelihood maximised above with $|T|$ replaced by $|T| + a - 1$ and $|H|$ by $|H| + b - 1$, so the same computation gives

$$\theta_{\mathrm{MAP}} = \frac{|T| + a - 1}{|H| + |T| + a + b - 2} \qquad (\text{for } |T| + a > 1,\ |H| + b > 1).$$

3.3 Full posterior and prediction

Normalising the posterior. θ_{MAP} is the most probable value of θ under the posterior, a single point. How certain is this estimate, and what is the probability that θ lies in some interval? Answering requires the entire posterior, not only its mode. By (1) the posterior is known up to a normalising constant, so the task is to find the constant that makes it integrate to 1. Computing $\int_0^1 \theta^{|T|+a-1}(1-\theta)^{|H|+b-1}\,\mathrm{d}\theta$ by brute force is tedious; it is easier to notice that the unnormalised posterior has the form of a Beta density.

Proposition 3.5 (Beta posterior). *With the prior $p(\theta) = \mathrm{Beta}(\theta \mid a, b)$ and the coin likelihood,*

$$p(\theta \mid \mathcal{D}) = \mathrm{Beta}(\theta \mid a + |T|, b + |H|) = \frac{\Gamma(|T| + a + |H| + b)}{\Gamma(|T| + a)\,\Gamma(|H| + b)}\, \theta^{|T|+a-1}(1-\theta)^{|H|+b-1}.$$

Proof. Let $\alpha = |T| + a$ and $\beta = |H| + b$. By (1), $p(\theta \mid \mathcal{D}) = c\,\theta^{\alpha-1}(1-\theta)^{\beta-1}$ for some constant c. Integrating over $[0,1]$ and using the normalisation of the Beta density (definition 3.4), $1 = c\,\frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$, so $c = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}$. □

The same trick solves any integral of a known density up to a constant factor: the integral is read off the normalising constant. For example, with the uniform prior $a = b = 1$ the 10 flips of section 3.1 give the posterior $\mathrm{Beta}(4,8) = 1320\,\theta^3(1-\theta)^7$, which is why $\int_0^1 \theta^3(1-\theta)^7\,\mathrm{d}\theta = \frac{1}{1320}$.

Definition 3.6 (Conjugate prior)**.** A prior is **conjugate** for a given likelihood if the posterior belongs to the same family of distributions as the prior.

The Beta distribution is conjugate for the Bernoulli likelihood, so the prior $\mathrm{Beta}(a,b)$ is not a lucky coincidence: the data only add $|T|$ to a and $|H|$ to b. Hence a and b can be read as numbers of tails and heads seen in the past.

Using the full posterior. The fully Bayesian approach gives an entire distribution, not just a point estimate (e.g. $\mathrm{Beta}(4,8)$ above, with mode 0.3). From it one can compute the expected value and the variance of θ under $p(\theta \mid \mathcal{D})$, or a **credible interval** $[\theta_1, \theta_2]$ with $\mathbb{P}(\theta \in [\theta_1, \theta_2] \mid \mathcal{D}) = 95\%$, not to be confused with a frequentist confidence interval.

Connections between the three approaches. The mode of $\mathrm{Beta}(\alpha,\beta)$ is $\frac{\alpha-1}{\alpha+\beta-2}$ for $\alpha,\beta > 1$ (the computation of θ_{MAP} in section 3.2 is exactly this maximisation). Hence θ_{MAP} is the mode of the posterior $\mathrm{Beta}(a+|T|, b+|H|)$, and with the uniform prior $a = b = 1$ (where $\mathrm{Beta}(\theta \mid 1,1) = 1$) it reduces to the MLE:

$$\theta_{\mathrm{MAP}} = \frac{|T| + 1 - 1}{|H| + |T| + 1 + 1 - 2} = \frac{|T|}{|H| + |T|} = \theta_{\mathrm{MLE}}.$$

All these closed formulas are a consequence of the conjugate prior; with a non-conjugate prior, $p(\theta \mid \mathcal{D})$ and θ_{MAP} need not have a closed form.

How many flips? With $a = b = 1$ the posterior is $\mathrm{Beta}(1+|T|, 1+|H|)$. For $(|T|,|H|) = (1,4), (10,40), (100,400)$ its mode stays at 0.2 but the density becomes more and more peaked (fig. 14): with more data we are more certain about θ.

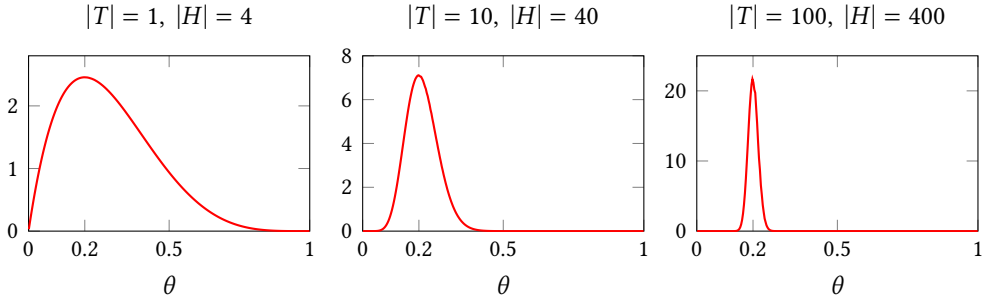


Figure 14: Posterior under the uniform prior for growing data.

From a frequentist point of view, $\theta_{\mathrm{MLE}} = \frac{|T|}{|T|+|H|} = 0.2$ for both $(1,4)$ and $(10,40)$, yet the second estimate is more reliable. A *sampling complexity bound* quantifies this.

Theorem 3.7 (Hoeffding's inequality)**.** *For $N = |T| + |H|$ i.i.d. flips of a coin with parameter θ and every $\varepsilon > 0$,*

$$\mathbb{P}\big(|\theta_{\mathrm{MLE}} - \theta| \geq \varepsilon\big) \leq 2e^{-2N\varepsilon^2}.$$

To know θ within ε with probability at least $1-\delta$, it suffices that $2e^{-2N\varepsilon^2} \le \delta$, i.e. (taking logarithms) $-2N\varepsilon^2 \le \ln\frac{\delta}{2}$, i.e.

$$N \ge \frac{\ln(2/\delta)}{2\varepsilon^2}.$$

For $\varepsilon = 0.1$ and $1-\delta = 0.99$ this gives $N \ge \frac{\ln 200}{0.02} \approx 264.9$, i.e. about 265 flips.

Predicting the next flip. The point estimates are plugged into the Bernoulli model: $p(F = \mathrm{T} \mid \theta_{\mathrm{MLE}}) = \mathrm{Ber}(F = \mathrm{T} \mid \theta_{\mathrm{MLE}}) = \theta_{\mathrm{MLE}}$, and in the same way $p(F = \mathrm{T} \mid \theta_{\mathrm{MAP}}) = \theta_{\mathrm{MAP}}$. With the entire posterior one can instead predict directly from the data and the prior. Write the outcome of the next flip as $f \in \{0, 1\}$, with 1 for tails, so that $p(f \mid \theta) = \mathrm{Ber}(f \mid \theta) = \theta^f(1-\theta)^{1-f}$.

Definition 3.8 (Posterior predictive distribution)**.** The **posterior predictive distribution** is the distribution of the next flip given the observations $\mathcal{D}$ and the prior parameters a, b,

$$p(F = f \mid \mathcal{D}, a, b) = p(f \mid \mathcal{D}, a, b),$$

not to be confused with the posterior $p(\theta \mid \mathcal{D}, a, b)$ over the parameter.

By the sum rule, the product rule and conditional independence,

$$\begin{aligned} p(f \mid \mathcal{D}, a, b) &= \int_0^1 p(f, \theta \mid \mathcal{D}, a, b)\,\mathrm{d}\theta = \int_0^1 p(f \mid \theta, \mathcal{D}, a, b)\, p(\theta \mid \mathcal{D}, a, b)\,\mathrm{d}\theta \\ &= \int_0^1 p(f \mid \theta)\, p(\theta \mid \mathcal{D}, a, b)\,\mathrm{d}\theta, \end{aligned}$$

where the last step uses that, once θ is known, the next flip is independent of the previous flips $\mathcal{D}$ (and of a, b). With the posterior $\mathrm{Beta}(\theta \mid \alpha, \beta)$ of proposition 3.5, $\alpha = |T| + a$, $\beta = |H| + b$, the case $f = 1$ is again an integral of a Beta density up to a constant:

$$\begin{aligned} p(f = 1 \mid \mathcal{D}, a, b) &= \int_0^1 \theta\ \mathrm{Beta}(\theta \mid \alpha, \beta)\,\mathrm{d}\theta = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \int_0^1 \theta^{\alpha}(1-\theta)^{\beta-1}\,\mathrm{d}\theta \\ &= \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \cdot \frac{\Gamma(\alpha+1)\,\Gamma(\beta)}{\Gamma(\alpha+\beta+1)} = \frac{\alpha}{\alpha+\beta}, \end{aligned}$$

using $\Gamma(x+1) = x\,\Gamma(x)$ (for $x = n \in \mathbb{N}$ this is $n! = n\,(n-1)!$); in the same way $p(f = 0 \mid \mathcal{D}, a, b) = \frac{\beta}{\alpha+\beta}$. Hence

$$p(f \mid \mathcal{D}, a, b) = \frac{(|T|+a)^f (|H|+b)^{1-f}}{|T|+a+|H|+b} = \mathrm{Ber}\left(f \,\middle|\, \frac{|T|+a}{|T|+a+|H|+b}\right).$$

The parameter θ no longer appears: it has been marginalised out. Predicting in this way is called **fully Bayesian analysis**.

Comparison. Table 2 summarises the three approaches for the coin.

	MLE	MAP	Fully Bayesian
Task	$\max_\theta \log p(\mathcal{D} \mid \theta)$	$\max_\theta \log p(\theta \mid \mathcal{D})$	find the normalising constant $p(\mathcal{D})$
Result	point estimate θ_{MLE}	point estimate θ_{MAP}	distribution $p(\theta \mid \mathcal{D})$
Coin	$\dfrac{\lvert T\rvert}{\lvert T\rvert + \lvert H\rvert}$	$\dfrac{\lvert T\rvert + a - 1}{\lvert T\rvert + \lvert H\rvert + a + b - 2}$	$\mathrm{Beta}(\theta \mid a + \lvert T\rvert, b + \lvert H\rvert)$
$p(F = \mathrm{T} \mid \cdot)$	$\dfrac{\lvert T\rvert}{\lvert T\rvert + \lvert H\rvert}$	$\dfrac{\lvert T\rvert + a - 1}{\lvert T\rvert + a + \lvert H\rvert + b - 2}$	$\dfrac{\lvert T\rvert + a}{\lvert T\rvert + a + \lvert H\rvert + b}$

Table 2: Parameter estimation and prediction for the coin.

Example 3.9. With the prior $a = b = 5$ and the counts $|T| = 4$, $|H| = 8$, the probability that the next flip is tails is

$$p_{\mathrm{MLE}} = \frac{4}{12} \approx 0.33, \qquad p_{\mathrm{MAP}} = \frac{4+5-1}{12+10-2} = \frac{8}{20} = 0.40, \qquad p_{\mathrm{FB}} = \frac{4+5}{12+10} = \frac{9}{22} \approx 0.41.$$

With $|T| = 304$, $|H| = 306$ instead, $p_{\mathrm{MLE}} = \frac{304}{610}$, $p_{\mathrm{MAP}} = \frac{308}{618}$ and $p_{\mathrm{FB}} = \frac{309}{620}$ are all ≈ 0.50: as more data are observed, the differences between the predictions become less noticeable.

4 Linear Regression

4.1 Basic linear regression

Regression model. Given observations $\mathcal{D} = \{(\boldsymbol{x}_i, y_i)\}_{i=1}^N$ with $\boldsymbol{x}_i \in \mathbb{R}^D$ and real-valued targets $y_i \in \mathbb{R}$, regression seeks a function f such that $y_i \approx f(\boldsymbol{x}_i)$, to predict the target of a new input. For instance, house areas x_i and prices y_i can be used to estimate the price of another house from its area. A model separates the systematic dependence from noise:

$$y_i = f(\boldsymbol{x}_i) + \epsilon_i, \qquad \epsilon_i \sim \mathcal{N}(0, \beta^{-1}), \tag{2}$$

where $\mathcal{N}$ denotes the normal distribution and $\beta > 0$ is the noise precision (inverse variance).

Definition 4.1 (Linear model and bias term)**.** A **linear regression model** has the form

$$f_{\boldsymbol{w}}(\boldsymbol{x}) = w_0 + \sum_{j=1}^{D} w_j x_j = w_0 + \boldsymbol{w}^T \boldsymbol{x}.$$

Here $\boldsymbol{w} = (w_1, \dots, w_D)^T$. The **bias** or **offset** w_0 is absorbed by augmenting the input and weight vectors:

$$\widetilde{\boldsymbol{x}} = (1, x_1, \dots, x_D)^T, \qquad \widetilde{\boldsymbol{w}} = (w_0, w_1, \dots, w_D)^T, \qquad f_{\boldsymbol{w}}(\boldsymbol{x}) = \widetilde{\boldsymbol{w}}^T \widetilde{\boldsymbol{x}}.$$

From now on the tildes are omitted: $\boldsymbol{x}$ and $\boldsymbol{w}$ include the offset, so $f_{\boldsymbol{w}}(\boldsymbol{x}) = \boldsymbol{w}^T \boldsymbol{x}$.

Least squares. A **loss function** measures the discrepancy between predictions and observed targets; fitting chooses weights that minimise it.

Definition 4.2 (Least-squares loss)**.** The **least-squares** (**LS**) loss is

$$E_{\mathrm{LS}}(\boldsymbol{w}) = \frac{1}{2} \sum_{i=1}^{N} \big(f_{\boldsymbol{w}}(\boldsymbol{x}_i) - y_i \big)^2 = \frac{1}{2} \sum_{i=1}^{N} (\boldsymbol{w}^T \boldsymbol{x}_i - y_i)^2. \tag{3}$$

Stacking the augmented inputs as rows of $\boldsymbol{X}$ and the targets into $\boldsymbol{y} = (y_1, \dots, y_N)^T$ gives

$$\boldsymbol{w}^* = \arg\min_{\boldsymbol{w}} E_{\mathrm{LS}}(\boldsymbol{w}) = \arg\min_{\boldsymbol{w}} \frac{1}{2} (\boldsymbol{X}\boldsymbol{w} - \boldsymbol{y})^T (\boldsymbol{X}\boldsymbol{w} - \boldsymbol{y}), \qquad \boldsymbol{X} \in \mathbb{R}^{N \times (D+1)}. \tag{4}$$

The factor $\frac{1}{2}$ simplifies differentiation and does not change the minimiser.

Proposition 4.3 (Normal equations)**.** *Least-squares minimisers solve the* ***normal equations***

$$\boldsymbol{X}^T \boldsymbol{X} \boldsymbol{w} = \boldsymbol{X}^T \boldsymbol{y}. \tag{5}$$

If $\boldsymbol{X}$ has full column rank, the minimiser is unique and

$$\boldsymbol{w}^* = (\boldsymbol{X}^T \boldsymbol{X})^{-1} \boldsymbol{X}^T \boldsymbol{y} = \boldsymbol{X}^\dagger \boldsymbol{y}. \tag{6}$$

Here $\boldsymbol{X}^\dagger$ is the ***Moore–Penrose pseudoinverse****; for an invertible square matrix it equals $\boldsymbol{X}^{-1}$. More generally, $\boldsymbol{X}^\dagger \boldsymbol{y}$ gives the least-squares solution of smallest L_2 norm, including when $\boldsymbol{X}^T \boldsymbol{X}$ is singular.*

Proof. Expand the quadratic, using $\boldsymbol{y}^T\boldsymbol{X}\boldsymbol{w} = \boldsymbol{w}^T\boldsymbol{X}^T\boldsymbol{y}$ because both are the same scalar:

$$E_{\mathrm{LS}}(\boldsymbol{w}) = \frac{1}{2}\big(\boldsymbol{w}^T\boldsymbol{X}^T\boldsymbol{X}\boldsymbol{w} - 2\boldsymbol{w}^T\boldsymbol{X}^T\boldsymbol{y} + \boldsymbol{y}^T\boldsymbol{y}\big).$$

Since $\boldsymbol{X}^T\boldsymbol{X}$ is symmetric, differentiating gives

$$\nabla_{\boldsymbol{w}} E_{\mathrm{LS}}(\boldsymbol{w}) = \boldsymbol{X}^T\boldsymbol{X}\boldsymbol{w} - \boldsymbol{X}^T\boldsymbol{y}, \qquad \nabla^2_{\boldsymbol{w}} E_{\mathrm{LS}} = \boldsymbol{X}^T\boldsymbol{X}.$$

For every vector $\boldsymbol{v}$,

$$\boldsymbol{v}^T\boldsymbol{X}^T\boldsymbol{X}\boldsymbol{v} = \|\boldsymbol{X}\boldsymbol{v}\|_2^2 \geq 0,$$

so the loss is convex and setting its gradient to zero gives its global minimisers, eq. (5). With full column rank the last expression is positive for $\boldsymbol{v} \neq \boldsymbol{0}$, hence the minimiser is unique; multiplying the normal equations by $(\boldsymbol{X}^T\boldsymbol{X})^{-1}$ gives eq. (6). □

Nonlinear dependence and basis functions. A straight line cannot capture, for example, data generated by $y_i = \sin(2\pi x_i) + \epsilon_i$ with $\epsilon_i \sim \mathcal{N}(0, \beta^{-1})$. For a scalar input, a polynomial instead uses

$$f_{\boldsymbol{w}}(x) = w_0 + \sum_{j=1}^{M} w_j x^j.$$

Increasing the degree allows approximation of any continuous function on a bounded closed interval to arbitrary accuracy. More generally, fixed transformations of the input provide nonlinear features while preserving linear dependence on the weights.

Definition 4.4 (Linear basis function model)**.** Here $\boldsymbol{x} \in \mathbb{R}^D$ again denotes the original, unaugmented feature vector. For **basis functions** $\phi_j : \mathbb{R}^D \to \mathbb{R}$, $j = 1, \ldots, M$, set $\phi_0(\boldsymbol{x}) = 1$ and $\boldsymbol{\phi}(\boldsymbol{x}) = (\phi_0(\boldsymbol{x}), \ldots, \phi_M(\boldsymbol{x}))^T$. The **linear basis function model** is

$$f_{\boldsymbol{w}}(\boldsymbol{x}) = w_0 + \sum_{j=1}^{M} w_j \phi_j(\boldsymbol{x}) = \boldsymbol{w}^T\boldsymbol{\phi}(\boldsymbol{x}). \tag{7}$$

It is linear in $\boldsymbol{w}$, even when nonlinear in $\boldsymbol{x}$. Its **design matrix** is

$$\boldsymbol{\Phi} = \begin{pmatrix} \phi_0(\boldsymbol{x}_1) & \phi_1(\boldsymbol{x}_1) & \cdots & \phi_M(\boldsymbol{x}_1) \\ \phi_0(\boldsymbol{x}_2) & \phi_1(\boldsymbol{x}_2) & \cdots & \phi_M(\boldsymbol{x}_2) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_0(\boldsymbol{x}_N) & \phi_1(\boldsymbol{x}_N) & \cdots & \phi_M(\boldsymbol{x}_N) \end{pmatrix} \in \mathbb{R}^{N \times (M+1)}.$$

Table 3 lists common choices for scalar inputs; in each case the constant basis function absorbs the offset.

Basis	$\phi_j(x)$, $j \geq 1$
Polynomial	x^j
Gaussian	$\exp\!\big(-\frac{(x-\mu_j)^2}{2s^2}\big)$
Logistic sigmoid	$\sigma\big(\frac{x-\mu_j}{s}\big)$, with $\sigma(a) = \frac{1}{1+e^{-a}}$

Table 3: Typical basis functions.

Here μ_j sets the location and $s > 0$ the width or scale.

With these features the same least-squares calculation applies, replacing $\boldsymbol{X}$ by $\boldsymbol{\Phi}$:

$$E_{\mathrm{LS}}(\boldsymbol{w}) = \frac{1}{2}\sum_{i=1}^{N}\bigl(\boldsymbol{w}^{T}\boldsymbol{\phi}(\boldsymbol{x}_i) - y_i\bigr)^2 = \frac{1}{2}(\boldsymbol{\Phi}\boldsymbol{w} - \boldsymbol{y})^{T}(\boldsymbol{\Phi}\boldsymbol{w} - \boldsymbol{y}). \tag{8}$$

Thus, by proposition 4.3,

$$\boldsymbol{w}^{*} = \boldsymbol{\Phi}^{\dagger}\boldsymbol{y} = (\boldsymbol{\Phi}^{T}\boldsymbol{\Phi})^{-1}\boldsymbol{\Phi}^{T}\boldsymbol{y} \quad \text{if } \boldsymbol{\Phi} \text{ has full column rank.} \tag{9}$$

Choosing the polynomial degree. The degree M controls the flexibility of a polynomial model. Figure 15 compares least-squares fits to ten noisy observations of $\sin(2\pi x)$ at $x_i = (i-1)/9$, $i = 1, \dots, 10$. Degrees 0 and 1 cannot represent the curvature (underfitting); degree 3 follows the main trend; degree 9 interpolates all ten points but oscillates sharply between them (overfitting).

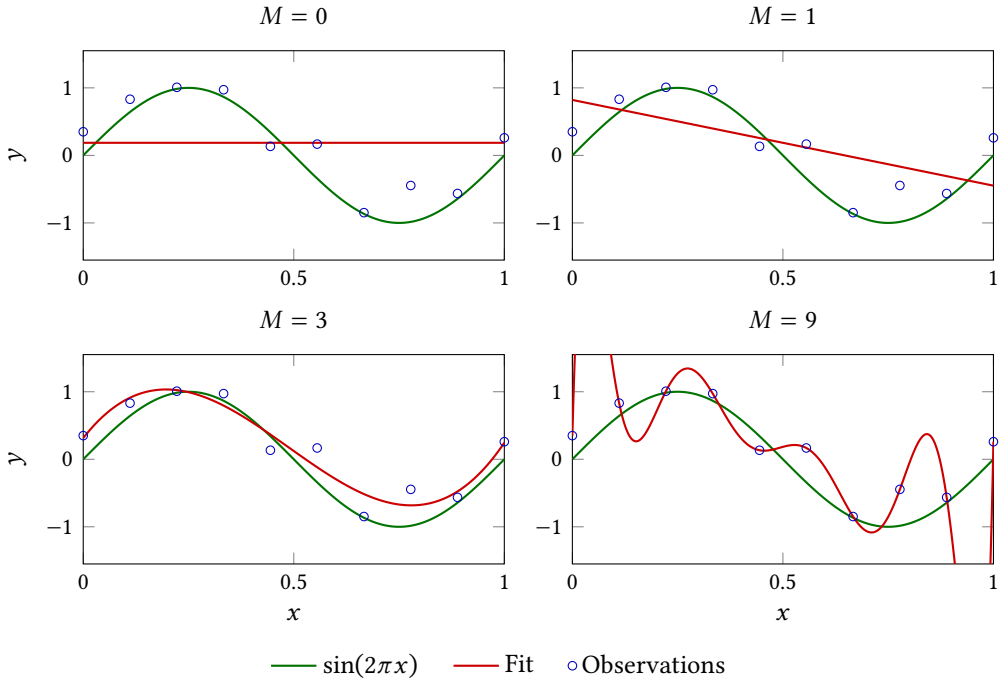


Figure 15: Polynomial least-squares fits to ten noisy observations.

Select M using the training–validation split of section 1.2, comparing errors through the **root mean squared error** (**RMSE**).

Definition 4.5 (Root mean squared error)**.** For N observations, the RMSE is

$$E_{\mathrm{RMS}} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\bigl(f_{\boldsymbol{w}}(\boldsymbol{x}_i) - y_i\bigr)^2} = \sqrt{\frac{2E_{\mathrm{LS}}(\boldsymbol{w})}{N}}.$$

In the polynomial experiment, training RMSE decreases to 0 at $M = 9$, while validation RMSE is minimal at $M = 6$ (about 0.38) and rises to about 1.3 at $M = 9$. Minimising the training error alone would therefore choose the interpolating model despite its poor generalisation.

Table 4 also shows that the high-degree interpolating fit has very large coefficients, whose terms must cancel to give target values of order 1.

Coefficient	$M=0$	$M=1$	$M=3$	$M=9$
w_0	0.19	0.82	0.31	0.35
w_1		−1.27	7.99	232.37
w_2			−25.43	−5321.83
w_3			17.37	48568.31
w_4				−231639.30
w_5				640042.26
w_6				−1061800.52
w_7				1042400.18
w_8				−557682.99
w_9				125201.43

Table 4: Polynomial coefficients rounded to two decimal places.

Regularisation. Penalising large weights provides another way to control overfitting without reducing the degree.

Definition 4.6 (Ridge regression)**. Ridge regression** adds an L_2 penalty to the least-squares loss:

$$E_{\text{ridge}}(\boldsymbol{w}) = \frac{1}{2}\sum_{i=1}^{N}\left(\boldsymbol{w}^T\boldsymbol{\phi}(\boldsymbol{x}_i) - y_i\right)^2 + \frac{\lambda}{2}\|\boldsymbol{w}\|_2^2, \qquad \lambda \geq 0, \tag{10}$$

where $\|\boldsymbol{w}\|_2^2 = \boldsymbol{w}^T\boldsymbol{w} = w_0^2 + w_1^2 + \cdots + w_M^2$ and λ is the **regularisation strength**. Here the penalty includes the offset w_0.

Larger λ favours a smaller weight norm. Its numerical value depends on whether the data loss is summed or averaged:

$$N\left(\frac{E_{\text{LS}}(\boldsymbol{w})}{N} + \frac{\lambda_{\text{mean}}}{2}\|\boldsymbol{w}\|_2^2\right) = E_{\text{LS}}(\boldsymbol{w}) + \frac{N\lambda_{\text{mean}}}{2}\|\boldsymbol{w}\|_2^2.$$

Thus the same fit uses $\lambda = N\lambda_{\text{mean}}$ in eq. (10). For the ten-point degree-9 model, mean-loss strengths 10^{-8} and 1 correspond to $\lambda = 10^{-7}$ and 10: the former suppresses the extreme oscillations, while the latter makes the fit too rigid to capture the sinusoid (fig. 16).

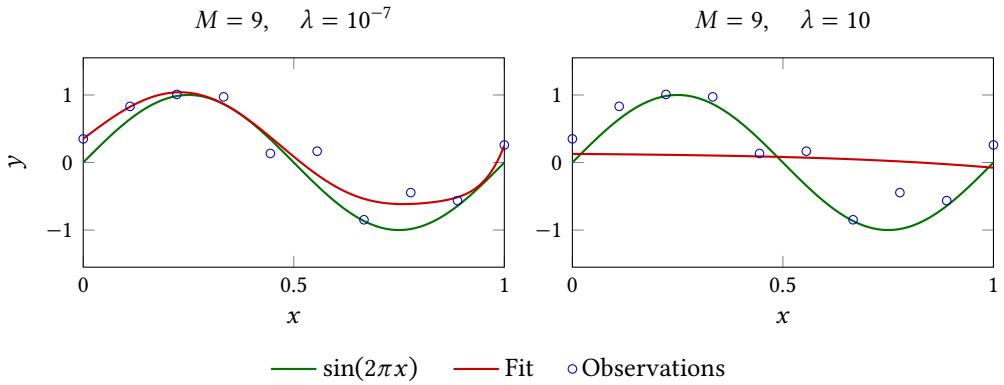


Figure 16: Degree-9 ridge fits with a summed data loss.

Bias and variance. **Bias** is the systematic discrepancy between the average fitted model and the true dependence; **variance** is the variation of the fitted model across random training sets. These refer to repeated fits, rather than the offset w_0. Figure 17 separates the two effects: the target centre is the truth

and each dot represents a fit from a different training set. For squared prediction error, bias and variance describe the reducible part; observation noise contributes an additional irreducible error.

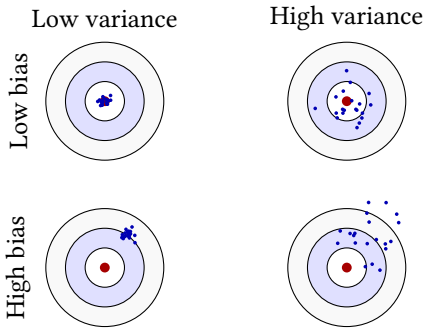


Figure 17: Bias and variance across training sets.

High bias occurs when a misspecified or overly rigid model cannot capture the underlying dependence, for example with excessive λ. High variance occurs when a flexible model follows noise in the training data, especially with high capacity and insufficient regularisation: this is overfitting. With unregularised least squares ($\lambda = 0$), increasing the polynomial degree reduces training error but eventually increases validation error, as in fig. 15. The **bias–variance trade-off** is that reducing one often increases the other; a common strategy is to choose a model with large capacity and control its variance by tuning λ on validation data.

Correlation and causation. In a linear model on the original features, w_j describes the change in the fitted target per unit change in feature j, holding the others fixed; its numerical size depends on the scales. For example, a least-squares fit between the number x of outdoor pool visitors and air temperature y (in °C) is

$$f(x) \approx 0.0197x + 15.41.$$

The small slope is compatible with a strong linear association because the two variables have very different units; after standardising both, the fitted slope is about 0.95. Correlation does not imply causation: adding visitors does not raise the air temperature. Temperature can also be a **confounding variable**, increasing both pool attendance and ice cream sales and thereby producing an association between them (fig. 18).

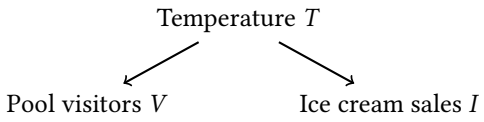


Figure 18: Temperature as a common cause.

4.2 Probabilistic estimation: maximum likelihood and MAP

Gaussian observation model. With the basis model of definition 4.4, the noise assumption in eq. (2) gives

$$p(y_i \mid \boldsymbol{x}_i, \boldsymbol{w}, \beta) = \mathcal{N}\left(y_i \mid \boldsymbol{w}^T \boldsymbol{\phi}(\boldsymbol{x}_i), \beta^{-1}\right).$$

The precision β is shared by all observations. Assume that the noise terms are independent, so the targets are conditionally independent given the inputs, weights and precision (fig. 19).

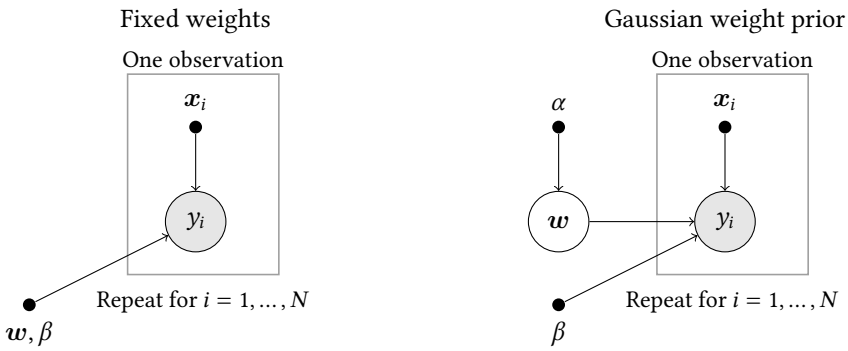


Figure 19: Fixed weights and a Gaussian weight prior.

In the diagrams, dots represent fixed quantities, shaded circles observed variables and the white circle an unknown random vector; each rectangle repeats its contents for $i = 1, \dots, N$.

Definition 4.7 (Gaussian regression likelihood)**.** Writing $\boldsymbol{X}$ for the collection of inputs and $\boldsymbol{\Phi}$ for their design matrix, the **likelihood** is

$$p(\boldsymbol{y} \mid \boldsymbol{X}, \boldsymbol{w}, \beta) = \prod_{i=1}^{N} \mathcal{N}\left(y_i \mid \boldsymbol{w}^T \boldsymbol{\phi}(\boldsymbol{x}_i), \beta^{-1}\right). \tag{11}$$

Its **negative log-likelihood** is $E_{\mathrm{ML}}(\boldsymbol{w}, \beta) = -\ln p(\boldsymbol{y} \mid \boldsymbol{X}, \boldsymbol{w}, \beta)$, and maximum likelihood estimation chooses

$$(\boldsymbol{w}_{\mathrm{ML}}, \beta_{\mathrm{ML}}) = \operatorname*{arg\,max}_{\boldsymbol{w},\, \beta>0} p(\boldsymbol{y} \mid \boldsymbol{X}, \boldsymbol{w}, \beta) = \operatorname*{arg\,min}_{\boldsymbol{w},\, \beta>0} E_{\mathrm{ML}}(\boldsymbol{w}, \beta),$$

because taking the logarithm preserves the maximisers.

Maximum likelihood and least squares.

Proposition 4.8 (Maximum likelihood estimates)**.** *For fixed $\beta > 0$, maximum likelihood weights minimise E_{LS} in eq. (8). In particular, $\boldsymbol{w}_{\mathrm{ML}} = \boldsymbol{\Phi}^{\dagger}\boldsymbol{y}$ is a solution, equal to $(\boldsymbol{\Phi}^T\boldsymbol{\Phi})^{-1}\boldsymbol{\Phi}^T\boldsymbol{y}$ when $\boldsymbol{\Phi}$ has full column rank. If the residual sum of squares is positive, the estimated noise variance is*

$$\beta_{\mathrm{ML}}^{-1} = \frac{1}{N}\sum_{i=1}^{N}\left(\boldsymbol{w}_{\mathrm{ML}}^T\boldsymbol{\phi}(\boldsymbol{x}_i) - y_i\right)^2 = \frac{2E_{\mathrm{LS}}(\boldsymbol{w}_{\mathrm{ML}})}{N}. \tag{12}$$

Proof. Substitute the Gaussian density into eq. (11) and turn the logarithm of a product into a sum:

$$\begin{aligned} E_{\mathrm{ML}}(\boldsymbol{w}, \beta) &= -\ln \prod_{i=1}^{N}\left[\sqrt{\frac{\beta}{2\pi}}\exp\left(-\frac{\beta}{2}\left(\boldsymbol{w}^T\boldsymbol{\phi}(\boldsymbol{x}_i) - y_i\right)^2\right)\right] \\ &= -\sum_{i=1}^{N}\left[\frac{1}{2}\ln\beta - \frac{1}{2}\ln(2\pi) - \frac{\beta}{2}\left(\boldsymbol{w}^T\boldsymbol{\phi}(\boldsymbol{x}_i) - y_i\right)^2\right] \\ &= \beta E_{\mathrm{LS}}(\boldsymbol{w}) - \frac{N}{2}\ln\beta + \frac{N}{2}\ln(2\pi). \end{aligned}$$

For fixed $\beta > 0$, the last two terms do not depend on $\boldsymbol{w}$ and multiplying the loss by β does not change its minimisers. Thus

$$\operatorname*{arg\,min}_{\boldsymbol{w}} E_{\mathrm{ML}}(\boldsymbol{w}, \beta) = \operatorname*{arg\,min}_{\boldsymbol{w}} E_{\mathrm{LS}}(\boldsymbol{w}),$$

whose solutions are given by proposition 4.3. Substitute such a solution and differentiate with respect to β:

$$\frac{\partial E_{\mathrm{ML}}}{\partial \beta} = E_{\mathrm{LS}}(\boldsymbol{w}_{\mathrm{ML}}) - \frac{N}{2\beta} = 0 \quad \Longrightarrow \quad \beta_{\mathrm{ML}}^{-1} = \frac{2E_{\mathrm{LS}}(\boldsymbol{w}_{\mathrm{ML}})}{N}.$$

The second derivative $N/(2\beta^2) > 0$ confirms the minimum. □

If all residuals are zero, the likelihood grows without bound as $\beta \to \infty$, so no finite joint MLE exists.

Gaussian prior and MAP. With little data, maximum likelihood can overfit. As for the coin in section 3.2, prior beliefs about the parameters can be incorporated by Bayes' formula. From here on, β is treated as known; uncertainty is assigned to $\boldsymbol{w}$, as in the right-hand diagram of fig. 19.

Definition 4.9 (Gaussian weight prior)**.** For $\alpha > 0$, use a zero-mean **isotropic Gaussian prior** on all $M+1$ weights:

$$p(\boldsymbol{w} \mid \alpha) = \mathcal{N}(\boldsymbol{w} \mid \boldsymbol{0}, \alpha^{-1}\boldsymbol{I}) = \left(\frac{\alpha}{2\pi}\right)^{(M+1)/2} \exp\left(-\frac{\alpha}{2}\boldsymbol{w}^T\boldsymbol{w}\right), \tag{13}$$

where α is the prior precision and $\boldsymbol{I} \in \mathbb{R}^{(M+1)\times(M+1)}$. Isotropy means equal variance in every direction. The prior favours small weight norms, discouraging overfitting, and its Gaussian form simplifies the posterior calculation.

Bayes' formula gives

$$p(\boldsymbol{w} \mid \boldsymbol{X}, \boldsymbol{y}, \alpha, \beta) = \frac{p(\boldsymbol{y} \mid \boldsymbol{X}, \boldsymbol{w}, \beta)\, p(\boldsymbol{w} \mid \alpha)}{p(\boldsymbol{y} \mid \boldsymbol{X}, \alpha, \beta)} \propto p(\boldsymbol{y} \mid \boldsymbol{X}, \boldsymbol{w}, \beta)\, p(\boldsymbol{w} \mid \alpha). \tag{14}$$

The denominator is the evidence and does not depend on $\boldsymbol{w}$. The roles of the coin parameter θ and likelihood $p(\mathcal{D} \mid \theta)$ are now played by $\boldsymbol{w}$ and $p(\boldsymbol{y} \mid \boldsymbol{X}, \boldsymbol{w}, \beta)$.

Proposition 4.10 (MAP and ridge regression)**.** *With the Gaussian prior, the MAP estimate is the ridge solution of definition 4.6 with $\lambda = \alpha/\beta$:*

$$\boldsymbol{w}_{\text{MAP}} = \arg\max_{\boldsymbol{w}} p(\boldsymbol{w} \mid \boldsymbol{X}, \boldsymbol{y}, \alpha, \beta) = \arg\min_{\boldsymbol{w}} \left[E_{\text{LS}}(\boldsymbol{w}) + \frac{\alpha}{2\beta}\|\boldsymbol{w}\|_2^2\right].$$

Proof. Taking the logarithm and dropping the log-evidence, which is constant in $\boldsymbol{w}$, reduces MAP estimation to minimising

$$\begin{aligned} E_{\text{MAP}}(\boldsymbol{w}) &= -\ln p(\boldsymbol{y} \mid \boldsymbol{X}, \boldsymbol{w}, \beta) - \ln p(\boldsymbol{w} \mid \alpha) \\ &= \beta E_{\text{LS}}(\boldsymbol{w}) - \frac{N}{2}\ln\beta + \frac{N}{2}\ln(2\pi) - \frac{M+1}{2}\ln\frac{\alpha}{2\pi} + \frac{\alpha}{2}\boldsymbol{w}^T\boldsymbol{w} \\ &= \beta\left[E_{\text{LS}}(\boldsymbol{w}) + \frac{\alpha}{2\beta}\|\boldsymbol{w}\|_2^2\right] + \text{const.} \end{aligned}$$

The constant and the positive factor β do not affect the minimiser, giving eq. (10) with $\lambda = \alpha/\beta$. □

4.3 Bayesian posterior and prediction

Full posterior over the weights. The fully Bayesian approach retains the entire posterior rather than only its mode. The Gaussian prior is conjugate to the Gaussian likelihood for the weights (definition 3.6). For fixed α, β, abbreviate the posterior conditioning by $\mathcal{D} = \{\boldsymbol{X}, \boldsymbol{y}\}$.

Proposition 4.11 (Gaussian weight posterior)**.** *The posterior is*

$$p(\boldsymbol{w} \mid \mathcal{D}) = \mathcal{N}(\boldsymbol{w} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma}), \qquad \boldsymbol{\Sigma}^{-1} = \alpha\boldsymbol{I} + \beta\boldsymbol{\Phi}^T\boldsymbol{\Phi}, \qquad \boldsymbol{\mu} = \beta\boldsymbol{\Sigma}\boldsymbol{\Phi}^T\boldsymbol{y}. \tag{15}$$

The parameters follow by collecting the quadratic and linear terms in the log-posterior and completing the square:

$$-2\ln p(\boldsymbol{w} \mid \mathcal{D}) = \boldsymbol{w}^T(\alpha\boldsymbol{I} + \beta\boldsymbol{\Phi}^T\boldsymbol{\Phi})\boldsymbol{w} - 2\beta\boldsymbol{w}^T\boldsymbol{\Phi}^T\boldsymbol{y} + \text{const} = (\boldsymbol{w} - \boldsymbol{\mu})^T\boldsymbol{\Sigma}^{-1}(\boldsymbol{w} - \boldsymbol{\mu}) + \text{const}.$$

The posterior mode equals its mean, so $\boldsymbol{w}_{\text{MAP}} = \boldsymbol{\mu}$. For $N = 0$, eq. (15) reduces to $\boldsymbol{\mu} = \boldsymbol{0}$ and $\boldsymbol{\Sigma} = \alpha^{-1}\boldsymbol{I}$, recovering the prior. Although the prior is isotropic, the posterior covariance is generally not diagonal: the observations couple the weights. With full column rank, an infinitely broad prior ($\alpha \to 0$) gives $\boldsymbol{w}_{\text{MAP}} \to \boldsymbol{w}_{\text{ML}}$.

Prediction from point estimates. For a new input $\boldsymbol{x}_{\text{new}}$, let $\boldsymbol{\phi}_{\text{new}} = \boldsymbol{\phi}(\boldsymbol{x}_{\text{new}})$. A **plug-in predictive distribution** substitutes a point estimate of the parameters into the observation model:

$$p(\hat{y}_{\text{new}} \mid \boldsymbol{x}_{\text{new}}, \boldsymbol{w}_{\text{ML}}, \beta_{\text{ML}}) = \mathcal{N}\left(\hat{y}_{\text{new}} \mid \boldsymbol{w}_{\text{ML}}^T \boldsymbol{\phi}_{\text{new}}, \beta_{\text{ML}}^{-1}\right), \tag{16}$$

$$p(\hat{y}_{\text{new}} \mid \boldsymbol{x}_{\text{new}}, \boldsymbol{w}_{\text{MAP}}, \beta) = \mathcal{N}\left(\hat{y}_{\text{new}} \mid \boldsymbol{w}_{\text{MAP}}^T \boldsymbol{\phi}_{\text{new}}, \beta^{-1}\right). \tag{17}$$

The first uses the ML noise estimate from eq. (12), while the second uses the fixed β assumed for Bayesian weight inference. Both variances are independent of the new input and account only for observation noise.

Integrating parameter uncertainty.

Definition 4.12 (Bayesian predictive distribution for regression)**.** The **posterior predictive distribution** averages the observation model over the weight posterior:

$$\begin{aligned} p(\hat{y}_{\text{new}} \mid \boldsymbol{x}_{\text{new}}, \boldsymbol{\mathcal{D}}) &= \int p(\hat{y}_{\text{new}}, \boldsymbol{w} \mid \boldsymbol{x}_{\text{new}}, \boldsymbol{\mathcal{D}})\, \mathrm{d}\boldsymbol{w} \\ &= \int p(\hat{y}_{\text{new}} \mid \boldsymbol{x}_{\text{new}}, \boldsymbol{w}, \beta)\, p(\boldsymbol{w} \mid \boldsymbol{\mathcal{D}})\, \mathrm{d}\boldsymbol{w} \qquad (18) \\ &= \mathcal{N}\left(\hat{y}_{\text{new}} \mid \boldsymbol{\mu}^T \boldsymbol{\phi}_{\text{new}}, \beta^{-1} + \boldsymbol{\phi}_{\text{new}}^T \boldsymbol{\Sigma} \boldsymbol{\phi}_{\text{new}}\right). \qquad (19) \end{aligned}$$

The factorisation in eq. (18) uses conditional independence: once the weights and noise precision are given, the new target does not depend on the training targets. A linear function of Gaussian weights is Gaussian, and independent Gaussian observation noise adds its variance:

$$\mathbb{E}[\hat{y}_{\text{new}} \mid \boldsymbol{x}_{\text{new}}, \boldsymbol{\mathcal{D}}] = \boldsymbol{\mu}^T \boldsymbol{\phi}_{\text{new}}, \qquad \operatorname{Var}(\hat{y}_{\text{new}} \mid \boldsymbol{x}_{\text{new}}, \boldsymbol{\mathcal{D}}) = \underbrace{\beta^{-1}}_{\text{observation noise}} + \underbrace{\boldsymbol{\phi}_{\text{new}}^T \boldsymbol{\Sigma} \boldsymbol{\phi}_{\text{new}}}_{\text{weight uncertainty}}.$$

The predictive mean agrees with the MAP prediction, but its variance includes uncertainty in the fitted weights and depends on the input. In the sinusoidal example $y_i = \sin(2\pi x_i) + \epsilon_i$, one or four observations leave broad uncertainty away from the observed inputs and allow very different posterior weight samples; with 25 observations spread over $[0, 1]$, the predictive mean follows the sinusoid and the uncertainty band is narrower.

5 Linear Classification

5.1 Linear decision boundaries

Classification problem. In regression the target is real-valued, e.g. the price of a house given its area (section 4); in **classification** it is one of C predetermined classes, e.g. whether a picture shows a cat or a dog, or which movement (rest, flexion, extension, co-contraction) produced two measured electrode voltages. Given observations $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_N \in \mathbb{R}^D$, stacked as the rows of the data matrix $\boldsymbol{X} \in \mathbb{R}^{N \times D}$, the set of classes $\mathcal{C} = \{1, \ldots, C\}$ and labels $\boldsymbol{y} = (y_1, \ldots, y_N)^T$ with $y_i \in \mathcal{C}$, the task is to find a function $f : \mathbb{R}^D \to \mathcal{C}$ with $y_i = f(\boldsymbol{x}_i)$ for $i = 1, \ldots, N$. Write $\hat{y}_i = f(\boldsymbol{x}_i)$ and $\hat{\boldsymbol{y}} = f(\boldsymbol{X})$ for the vector of predictions.

Definition 5.1 (Zero-one loss)**.** The **zero-one loss** counts the misclassified samples:

$$\ell_{01}(\boldsymbol{y}, \hat{\boldsymbol{y}}) = \sum_{i=1}^{N} \mathbb{I}(\hat{y}_i \neq y_i).$$

Hyperplanes. For two classes, $C = \{0, 1\}$, the simplest way to separate the points of the two classes is a **hyperplane**

$$\{\boldsymbol{x} \in \mathbb{R}^D \;:\; \boldsymbol{w}^T\boldsymbol{x} + w_0 = 0\},$$

defined by a **normal vector** $\boldsymbol{w} \neq \mathbf{0}$ and an **offset** w_0. The sign of $\boldsymbol{w}^T\boldsymbol{x} + w_0$ tells on which side $\boldsymbol{x}$ lies: it is positive on the side $\boldsymbol{w}$ points to, negative on the other side and zero on the hyperplane. Hyperplanes are computationally convenient: they are easy to evaluate. In fig. 20, $r = (\boldsymbol{w}^T\boldsymbol{x} + w_0)/\|\boldsymbol{w}\|$ is the signed distance of $\boldsymbol{x}$ from the hyperplane, because writing $\boldsymbol{x} = \boldsymbol{x}_\perp + r\,\boldsymbol{w}/\|\boldsymbol{w}\|$, with $\boldsymbol{x}_\perp$ the orthogonal projection onto the hyperplane, gives $\boldsymbol{w}^T\boldsymbol{x} + w_0 = \boldsymbol{w}^T\boldsymbol{x}_\perp + w_0 + r\|\boldsymbol{w}\| = r\|\boldsymbol{w}\|$; for $\boldsymbol{x} = \mathbf{0}$ this shows that the hyperplane lies at signed distance $-w_0/\|\boldsymbol{w}\|$ from the origin along $\boldsymbol{w}$.

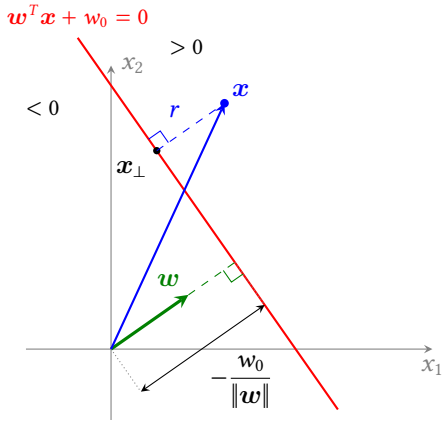


Figure 20: Geometry of a hyperplane in $\mathbb{R}^2$.

Definition 5.2 (Linear separability)**.** A dataset $\mathcal{D} = \{(\boldsymbol{x}_i, y_i)\}_{i=1}^N$ with $y_i \in \{0, 1\}$ is **linearly separable** if there is a hyperplane with all $\boldsymbol{x}_i$ with $y_i = 0$ on one side and all $\boldsymbol{x}_i$ with $y_i = 1$ on the other side.

Perceptron.

Definition 5.3 (Perceptron)**.** The **perceptron**, one of the oldest methods for binary classification, predicts

$$\hat{y} = f(\boldsymbol{w}^T\boldsymbol{x} + w_0), \qquad f(t) = \begin{cases} 1 & \text{if } t > 0, \\ 0 & \text{otherwise,} \end{cases}$$

where f is the **step function** (fig. 23).

The parameters are learned from the training set as follows:

1. initialise $\boldsymbol{w}$ and w_0 to any value, e.g. $\boldsymbol{w} = \mathbf{0}$, $w_0 = 0$;
2. for each misclassified training sample $\boldsymbol{x}_i$, update

$$\boldsymbol{w} \leftarrow \boldsymbol{w} + \boldsymbol{x}_i, \quad w_0 \leftarrow w_0 + 1 \quad \text{if } y_i = 1; \qquad \boldsymbol{w} \leftarrow \boldsymbol{w} - \boldsymbol{x}_i, \quad w_0 \leftarrow w_0 - 1 \quad \text{if } y_i = 0;$$

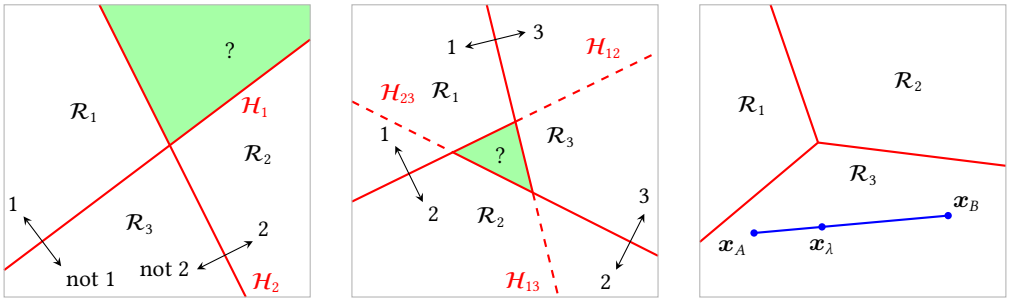
3. repeat step 2 until all training samples are classified correctly.

Theorem 5.4 (Perceptron convergence)**.** *If the training set is linearly separable, the perceptron learning rule stops after a finite number of updates, with a hyperplane* $(\boldsymbol{w}, w_0)$ *that separates the two classes.*

The number of updates needed cannot be determined in advance.

More than two classes. Combining two-class hyperplanes does not extend cleanly to $C > 2$ classes, because some regions remain ambiguous (fig. 21):

- a **one-versus-rest** classifier uses hyperplanes $\mathcal{H}_c$, each separating class c from all other classes. With three classes and two hyperplanes $\mathcal{H}_1, \mathcal{H}_2$ (class 3 where both answer "not"), the region that both classes 1 and 2 claim is ambiguous;
- a **one-versus-one** classifier uses a hyperplane $\mathcal{H}_{cc'}$ for each pair of classes $c \neq c'$, deciding between c and c', and classifies by majority vote. With three classes, the vote is tied in the triangle where every class wins exactly one of its two pairwise decisions.



(a) one-versus-rest (b) one-versus-one (c) multiclass discriminant

Figure 21: Linear classifiers for three classes; ambiguous regions shaded.

Definition 5.5 (Multiclass linear discriminant)**.** Take C linear functions $f_c(\boldsymbol{x}) = \boldsymbol{w}_c^T \boldsymbol{x} + w_{c0}$, $c \in \mathcal{C}$, and assign $\boldsymbol{x}$ to the class with the highest value:

$$\hat{y} = \arg\max_{c \in \mathcal{C}} f_c(\boldsymbol{x}).$$

This divides the domain into convex decision regions $\mathcal{R}_c = \{\boldsymbol{x} : \hat{y} = c\}$ (fig. 21(c)): if $\boldsymbol{x}_A, \boldsymbol{x}_B \in \mathcal{R}_c$, then every $\boldsymbol{x}_\lambda = \lambda \boldsymbol{x}_A + (1-\lambda)\boldsymbol{x}_B$ with $\lambda \in [0, 1]$ is in $\mathcal{R}_c$ too, because each $f_{c'}$ is affine, so the inequalities $f_c \geq f_{c'}$ at $\boldsymbol{x}_A$ and $\boldsymbol{x}_B$ carry over to $\boldsymbol{x}_\lambda$.

Nonlinear boundaries. If the classes are not linearly separable, e.g. class 0 points scattered around the origin and class 1 points on a ring around them, no hyperplane separates them. As in regression (definition 4.4), a fixed nonlinear transformation $\boldsymbol{\phi} : \mathbb{R}^D \to \mathbb{R}^M$ can map the samples to a space where they are linearly separable; a linear classifier is then applied to $\boldsymbol{\phi}(\boldsymbol{x})$. For the ring, the polar coordinates

$$\boldsymbol{\phi}(\boldsymbol{x}) = (\theta, r) = \big(\text{angle}(\boldsymbol{x}), \|\boldsymbol{x}\|_2\big)$$

make the classes separable by a line $r = r_0$, on whose two sides $r - r_0$ has opposite signs (fig. 22). These basis functions are fixed in advance; kernels and adaptive (learned) basis functions, treated in later lectures, develop the idea further.

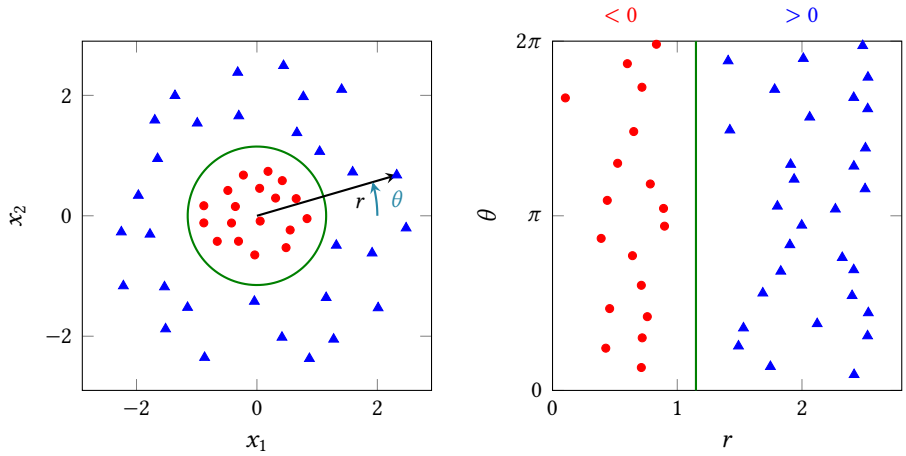


Figure 22: Polar basis functions for two classes.

Limitations of hard decisions. The classifiers above make hard decisions: they output a label and nothing else. They give no measure of uncertainty, cannot handle noisy data, generalise poorly and are difficult to optimise.

Probabilistic classifiers. The alternative is to model the distribution of the label y given the data $\boldsymbol{x}$. By Bayes' formula (definition 3.2),

$$p(y = c \mid \boldsymbol{x}) = \frac{p(\boldsymbol{x} \mid y = c)\, p(y = c)}{p(\boldsymbol{x})}. \tag{20}$$

A **generative** model specifies the joint distribution $p(\boldsymbol{x}, y = c) = p(\boldsymbol{x} \mid y = c)\, p(y = c)$ and obtains $p(y = c \mid \boldsymbol{x})$ from eq. (20) (section 5.2); a **discriminative** model specifies $p(y = c \mid \boldsymbol{x})$ directly (section 5.3). Given $p(y \mid \boldsymbol{x})$, the prediction $\hat{y}$ can be chosen to suit the problem; a popular choice is the mode, $\hat{y} = \arg\max_{c \in C} p(y = c \mid \boldsymbol{x})$.

5.2 Generative models: LDA and naive Bayes

Generative classifiers. In eq. (20), the **class prior** $p(y = c)$ is the probability that a point belongs to class c before seeing it, and the **class conditional** $p(\boldsymbol{x} \mid y = c)$ is the distribution of the points generated by class c. The evidence $p(\boldsymbol{x})$ does not depend on c, so $p(y = c \mid \boldsymbol{x}) \propto p(\boldsymbol{x} \mid y = c)\, p(y = c)$. A generative classifier is used as follows. First choose parametric models $p(\boldsymbol{x} \mid y = c, \boldsymbol{\psi})$ and $p(y = c \mid \boldsymbol{\theta})$. In **learning**, estimate the parameters $\{\boldsymbol{\psi}, \boldsymbol{\theta}\}$ from the data $\mathcal{D}$, e.g. by maximum likelihood, obtaining $\{\hat{\boldsymbol{\psi}}, \hat{\boldsymbol{\theta}}\}$. In **inference**, classify a new $\boldsymbol{x}$ by Bayes' formula,

$$p(y = c \mid \boldsymbol{x}, \hat{\boldsymbol{\psi}}, \hat{\boldsymbol{\theta}}) \propto p(\boldsymbol{x} \mid y = c, \hat{\boldsymbol{\psi}})\, p(y = c \mid \hat{\boldsymbol{\theta}}).$$

The fitted model can also **generate** new data, hence its name: sample a label $y_{\text{new}} \sim p(y \mid \hat{\boldsymbol{\theta}})$, then a feature vector $\boldsymbol{x}_{\text{new}} \sim p(\boldsymbol{x} \mid y = y_{\text{new}}, \hat{\boldsymbol{\psi}})$. From now on the conditioning on the parameters is left implicit.

Class prior.

Definition 5.6 (Categorical class prior)**.** The label takes one of C discrete values, so it is modelled by a **categorical distribution**, $y \sim \text{Categorical}(\boldsymbol{\theta})$:

$$p(y = c) = \theta_c, \qquad \text{equivalently} \qquad p(y) = \prod_{c=1}^{C} \theta_c^{\mathbb{I}(y=c)},$$

where the parameter $\boldsymbol{\theta} \in \mathbb{R}^C$ satisfies $0 \le \theta_c \le 1$ and $\sum_{c=1}^{C} \theta_c = 1$.

Proposition 5.7 (MLE of the class prior)**.** *Let $N_c = \sum_{i=1}^{N} \mathbb{I}(y_i = c)$ be the number of training samples of class c. The maximum likelihood estimate of $\boldsymbol{\theta}$ is the class frequency*

$$\theta_c^{\text{MLE}} = \frac{1}{N} \sum_{i=1}^{N} \mathbb{I}(y_i = c) = \frac{N_c}{N}.$$

Gaussian class conditionals.

Definition 5.8 (Gaussian class conditionals)**.** For continuous features $\boldsymbol{x} \in \mathbb{R}^D$, each class conditional is a multivariate normal distribution with a class mean $\boldsymbol{\mu}_c$ and a covariance matrix $\boldsymbol{\Sigma}$ shared by all classes:

$$p(\boldsymbol{x} \mid y = c) = \mathcal{N}(\boldsymbol{x} \mid \boldsymbol{\mu}_c, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{D/2}|\boldsymbol{\Sigma}|^{1/2}} \exp\Big(-\frac{1}{2}(\boldsymbol{x} - \boldsymbol{\mu}_c)^T \boldsymbol{\Sigma}^{-1}(\boldsymbol{x} - \boldsymbol{\mu}_c)\Big), \tag{21}$$

where $|\boldsymbol{\Sigma}|$ is the determinant of $\boldsymbol{\Sigma}$.

The covariance is shared because estimating a separate $\boldsymbol{\Sigma}_c$ for each class behaves badly numerically, unless a lot of data is available.

Proposition 5.9 (MLE of the Gaussian class conditionals)**.** *With the categorical prior and eq. (21), the maximum likelihood estimates are the class frequencies of proposition 5.7, the class means and the* ***pooled covariance***

$$\hat{\boldsymbol{\mu}}_c = \frac{1}{N_c} \sum_{i:\, y_i = c} \boldsymbol{x}_i, \qquad \hat{\boldsymbol{\Sigma}} = \frac{1}{N} \sum_{i=1}^{N} (\boldsymbol{x}_i - \hat{\boldsymbol{\mu}}_{y_i})(\boldsymbol{x}_i - \hat{\boldsymbol{\mu}}_{y_i})^T = \sum_{c=1}^{C} \frac{N_c}{N} \boldsymbol{S}_c,$$

where $\boldsymbol{S}_c = \frac{1}{N_c} \sum_{i:\, y_i = c} (\boldsymbol{x}_i - \hat{\boldsymbol{\mu}}_c)(\boldsymbol{x}_i - \hat{\boldsymbol{\mu}}_c)^T$ is the covariance of class c around its mean.

Proof. For i.i.d. samples the likelihood is $\prod_i p(y_i)\, p(\boldsymbol{x}_i \mid y_i)$, so its logarithm splits into $\sum_i \log \theta_{y_i}$, maximised by proposition 5.7, and, up to the constant $-\frac{ND}{2}\log(2\pi)$,

$$\ell(\boldsymbol{\mu}_1, \dots, \boldsymbol{\mu}_C, \boldsymbol{\Sigma}) = -\frac{N}{2}\log|\boldsymbol{\Sigma}| - \frac{1}{2}\sum_{i=1}^{N} (\boldsymbol{x}_i - \boldsymbol{\mu}_{y_i})^T \boldsymbol{\Sigma}^{-1} (\boldsymbol{x}_i - \boldsymbol{\mu}_{y_i}).$$

Only the samples of class c involve $\boldsymbol{\mu}_c$. Since $\boldsymbol{\Sigma}^{-1}$ is symmetric,

$$\nabla_{\boldsymbol{\mu}_c} \ell = \boldsymbol{\Sigma}^{-1} \sum_{i:\, y_i = c} (\boldsymbol{x}_i - \boldsymbol{\mu}_c) = \boldsymbol{0} \quad \Longleftrightarrow \quad \boldsymbol{\mu}_c = \frac{1}{N_c} \sum_{i:\, y_i = c} \boldsymbol{x}_i,$$

a maximum because ℓ is a concave quadratic function of $\boldsymbol{\mu}_c$; the result does not depend on $\boldsymbol{\Sigma}$. For the covariance, write $\boldsymbol{\Lambda} = \boldsymbol{\Sigma}^{-1}$, use $\log|\boldsymbol{\Sigma}| = -\log|\boldsymbol{\Lambda}|$ and $\boldsymbol{a}^T \boldsymbol{\Lambda} \boldsymbol{a} = \operatorname{tr}(\boldsymbol{\Lambda} \boldsymbol{a}\boldsymbol{a}^T)$, and insert the means $\hat{\boldsymbol{\mu}}_c$:

$$\ell = \frac{N}{2}\log|\boldsymbol{\Lambda}| - \frac{1}{2}\operatorname{tr}(\boldsymbol{\Lambda}\boldsymbol{S}), \qquad \boldsymbol{S} = \sum_{i=1}^{N} (\boldsymbol{x}_i - \hat{\boldsymbol{\mu}}_{y_i})(\boldsymbol{x}_i - \hat{\boldsymbol{\mu}}_{y_i})^T.$$

With the matrix derivatives $\nabla_{\boldsymbol{\Lambda}} \log|\boldsymbol{\Lambda}| = \boldsymbol{\Lambda}^{-1}$ and $\nabla_{\boldsymbol{\Lambda}} \operatorname{tr}(\boldsymbol{\Lambda}\boldsymbol{S}) = \boldsymbol{S}$ (for symmetric $\boldsymbol{\Lambda}$ and $\boldsymbol{S}$),

$$\nabla_{\boldsymbol{\Lambda}} \ell = \frac{N}{2}\boldsymbol{\Lambda}^{-1} - \frac{1}{2}\boldsymbol{S} = \boldsymbol{0} \quad \Longleftrightarrow \quad \boldsymbol{\Sigma} = \boldsymbol{\Lambda}^{-1} = \frac{1}{N}\boldsymbol{S},$$

a maximum because $\log|\boldsymbol{\Lambda}|$ is concave and the trace is linear in $\boldsymbol{\Lambda}$. Grouping the terms of $\boldsymbol{S}$ by class gives $\boldsymbol{S} = \sum_c N_c \boldsymbol{S}_c$. □

Posterior as a sigmoid. For two classes, $C = \{0, 1\}$, expand $p(\boldsymbol{x})$ in eq. (20) by the sum rule and divide numerator and denominator by $p(\boldsymbol{x} \mid y = 1)\, p(y = 1)$:

$$p(y = 1 \mid \boldsymbol{x}) = \frac{p(\boldsymbol{x} \mid y = 1)\, p(y = 1)}{p(\boldsymbol{x} \mid y = 1)\, p(y = 1) + p(\boldsymbol{x} \mid y = 0)\, p(y = 0)} = \frac{1}{1 + \exp(-a)} = \sigma(a), \tag{22}$$

where

$$a = \log \frac{p(\boldsymbol{x} \mid y = 1)\, p(y = 1)}{p(\boldsymbol{x} \mid y = 0)\, p(y = 0)} \tag{23}$$

and $\sigma(a) = 1/(1 + e^{-a})$ is the logistic **sigmoid function** of table 3, plotted in fig. 23 together with the step function of the perceptron.

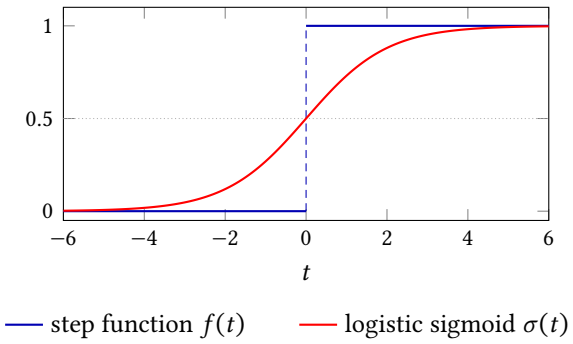


Figure 23: Step function and logistic sigmoid.

Linear discriminant analysis.

Proposition 5.10 (LDA for two classes)**.** *With Gaussian class conditionals sharing the covariance* $\boldsymbol{\Sigma}$ *(definition 5.8), the posterior is a sigmoid of a linear function of* $\boldsymbol{x}$*,*

$$p(y = 1 \mid \boldsymbol{x}) = \sigma(\boldsymbol{w}^T \boldsymbol{x} + w_0), \qquad \textit{i.e.} \qquad y \mid \boldsymbol{x} \sim \mathrm{Ber}\big(\sigma(\boldsymbol{w}^T \boldsymbol{x} + w_0)\big), \tag{24}$$

with

$$\boldsymbol{w} = \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0), \qquad w_0 = -\frac{1}{2}\boldsymbol{\mu}_1^T \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_1 + \frac{1}{2}\boldsymbol{\mu}_0^T \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_0 + \log \frac{p(y = 1)}{p(y = 0)}. \tag{25}$$

This classifier is **linear discriminant analysis** (**LDA**).

Proof. By eq. (22) it suffices to show that $a = \boldsymbol{w}^T \boldsymbol{x} + w_0$. The normalising constant $(2\pi)^{-D/2} |\boldsymbol{\Sigma}|^{-1/2}$ is the same for both classes and cancels in eq. (23), so

$$a = -\frac{1}{2}(\boldsymbol{x} - \boldsymbol{\mu}_1)^T \boldsymbol{\Sigma}^{-1} (\boldsymbol{x} - \boldsymbol{\mu}_1) + \frac{1}{2}(\boldsymbol{x} - \boldsymbol{\mu}_0)^T \boldsymbol{\Sigma}^{-1} (\boldsymbol{x} - \boldsymbol{\mu}_0) + \log \frac{p(y = 1)}{p(y = 0)}.$$

Since $\boldsymbol{\Sigma}^{-1}$ is symmetric, $\boldsymbol{x}^T \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_c = \boldsymbol{\mu}_c^T \boldsymbol{\Sigma}^{-1} \boldsymbol{x}$ and each quadratic form expands as

$$(\boldsymbol{x} - \boldsymbol{\mu}_c)^T \boldsymbol{\Sigma}^{-1} (\boldsymbol{x} - \boldsymbol{\mu}_c) = \boldsymbol{x}^T \boldsymbol{\Sigma}^{-1} \boldsymbol{x} - 2\boldsymbol{\mu}_c^T \boldsymbol{\Sigma}^{-1} \boldsymbol{x} + \boldsymbol{\mu}_c^T \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_c.$$

The terms $\boldsymbol{x}^T \boldsymbol{\Sigma}^{-1} \boldsymbol{x}$ cancel, leaving

$$a = (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)^T \boldsymbol{\Sigma}^{-1} \boldsymbol{x} - \frac{1}{2}\boldsymbol{\mu}_1^T \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_1 + \frac{1}{2}\boldsymbol{\mu}_0^T \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_0 + \log \frac{p(y = 1)}{p(y = 0)} = \boldsymbol{w}^T \boldsymbol{x} + w_0,$$

because $(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)^T \boldsymbol{\Sigma}^{-1} = \big(\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)\big)^T$. □

Hence LDA leads to a linear decision boundary, the hyperplane $\boldsymbol{w}^T\boldsymbol{x}+w_0 = 0$ where $p(y=1 \mid \boldsymbol{x}) = \frac{1}{2}$. Figure 24 shows the joint densities $p(x, y=c)$ and the posterior for $D=1$, $\mu_0 = -1$, $\mu_1 = 1$, $\Sigma = 1$ and $p(y=1) = 0.3$, for which $w = 2$ and $w_0 = \log\frac{3}{7}$.

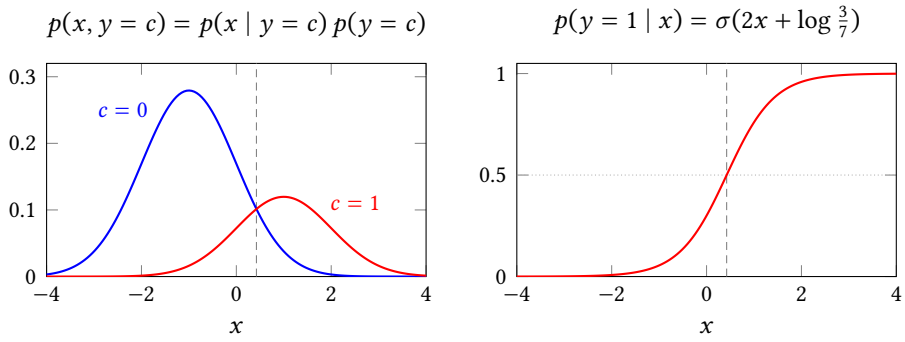


Figure 24: LDA in one dimension.

More than two classes.

Definition 5.11 (Softmax)**.** The **softmax function** $\sigma : \mathbb{R}^K \to \Delta^{K-1}$ is

$$\sigma(\boldsymbol{a})_j = \frac{\exp(a_j)}{\sum_{k=1}^K \exp(a_k)}, \qquad j = 1, \dots, K,$$

where $\Delta^{K-1} = \left\{ \boldsymbol{p} \in \mathbb{R}^K \;:\; \sum_{k=1}^K p_k = 1,\ p_k \geq 0 \text{ for } k = 1, \dots, K \right\}$ is the standard **probability simplex**.

The softmax generalises the sigmoid to K dimensions.

Proposition 5.12 (LDA for more than two classes)**.** *With Gaussian class conditionals sharing* $\boldsymbol{\Sigma}$*,*

$$p(y=c \mid \boldsymbol{x}) = \frac{p(\boldsymbol{x} \mid y=c)\, p(y=c)}{\sum_{c'=1}^C p(\boldsymbol{x} \mid y=c')\, p(y=c')} = \frac{\exp(\boldsymbol{w}_c^T\boldsymbol{x} + w_{c0})}{\sum_{c'=1}^C \exp(\boldsymbol{w}_{c'}^T\boldsymbol{x} + w_{c'0})},$$

the softmax of $(\boldsymbol{w}_1^T\boldsymbol{x} + w_{10}, \dots, \boldsymbol{w}_C^T\boldsymbol{x} + w_{C0})$*, with*

$$\boldsymbol{w}_c = \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_c, \qquad w_{c0} = -\frac{1}{2}\boldsymbol{\mu}_c^T\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_c + \log p(y=c).$$

Proof. Expanding the quadratic form as in the proof of proposition 5.10,

$$\begin{aligned} p(\boldsymbol{x} \mid y=c)\, p(y=c) &= \underbrace{\frac{\exp\left(-\frac{1}{2}\boldsymbol{x}^T\boldsymbol{\Sigma}^{-1}\boldsymbol{x}\right)}{(2\pi)^{D/2}|\boldsymbol{\Sigma}|^{1/2}}}_{g(\boldsymbol{x})} \exp\left(\boldsymbol{\mu}_c^T\boldsymbol{\Sigma}^{-1}\boldsymbol{x} - \frac{1}{2}\boldsymbol{\mu}_c^T\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_c + \log p(y=c)\right) \\ &= g(\boldsymbol{x}) \exp(\boldsymbol{w}_c^T\boldsymbol{x} + w_{c0}). \end{aligned}$$

The factor $g(\boldsymbol{x})$ does not depend on c, so it cancels between numerator and denominator. □

Naive Bayes.

Definition 5.13 (Naive Bayes)**. Naive Bayes** assumes that the D features of $\boldsymbol{x} = (x_1, \dots, x_D)^T$ are conditionally independent given the class:

$$p(x_1, \dots, x_D \mid y=c) = \prod_{j=1}^D p(x_j \mid y=c).$$

For continuous features with Gaussian factors $p(x_j \mid y = c) = \mathcal{N}(x_j \mid \mu_{cj}, \sigma_{cj}^2)$, this corresponds to a Gaussian class conditional $\mathcal{N}(\boldsymbol{x} \mid \boldsymbol{\mu}_c, \boldsymbol{\Sigma}_c)$ with the *diagonal* covariance matrix $\boldsymbol{\Sigma}_c = \operatorname{diag}(\sigma_{c1}^2, \dots, \sigma_{cD}^2)$. Unlike LDA, where all classes share the same covariance matrix $\boldsymbol{\Sigma}$, naive Bayes uses a different covariance matrix $\boldsymbol{\Sigma}_c$ for each class.

Proposition 5.14 (Quadratic decision boundary)**.** *For two classes with class conditionals $\mathcal{N}(\boldsymbol{x} \mid \boldsymbol{\mu}_c, \boldsymbol{\Sigma}_c)$ and priors $\pi_c = p(y = c)$, the quantity a of eq. (23) is*

$$a = \boldsymbol{x}^T \boldsymbol{W}_2 \boldsymbol{x} + \boldsymbol{w}_1^T \boldsymbol{x} + w_0,$$

with

$$\boldsymbol{W}_2 = \frac{1}{2}\left(\boldsymbol{\Sigma}_0^{-1} - \boldsymbol{\Sigma}_1^{-1}\right), \qquad \boldsymbol{w}_1 = \boldsymbol{\Sigma}_1^{-1}\boldsymbol{\mu}_1 - \boldsymbol{\Sigma}_0^{-1}\boldsymbol{\mu}_0,$$
$$w_0 = -\frac{1}{2}\boldsymbol{\mu}_1^T\boldsymbol{\Sigma}_1^{-1}\boldsymbol{\mu}_1 + \frac{1}{2}\boldsymbol{\mu}_0^T\boldsymbol{\Sigma}_0^{-1}\boldsymbol{\mu}_0 + \log\frac{\pi_1}{\pi_0} + \frac{1}{2}\log\frac{|\boldsymbol{\Sigma}_0|}{|\boldsymbol{\Sigma}_1|}.$$

Proof. Now the normalising constants differ between the classes and the quadratic terms do not cancel. Expanding as in the proof of proposition 5.10,

$$\log \mathcal{N}(\boldsymbol{x} \mid \boldsymbol{\mu}_c, \boldsymbol{\Sigma}_c) = -\frac{D}{2}\log(2\pi) - \frac{1}{2}\log|\boldsymbol{\Sigma}_c| - \frac{1}{2}\boldsymbol{x}^T\boldsymbol{\Sigma}_c^{-1}\boldsymbol{x} + \boldsymbol{\mu}_c^T\boldsymbol{\Sigma}_c^{-1}\boldsymbol{x} - \frac{1}{2}\boldsymbol{\mu}_c^T\boldsymbol{\Sigma}_c^{-1}\boldsymbol{\mu}_c.$$

Subtracting this expression for $c = 0$ from the one for $c = 1$ and adding $\log(\pi_1/\pi_0)$ gives the quadratic term $\frac{1}{2}\boldsymbol{x}^T(\boldsymbol{\Sigma}_0^{-1} - \boldsymbol{\Sigma}_1^{-1})\boldsymbol{x}$, the linear term $(\boldsymbol{\Sigma}_1^{-1}\boldsymbol{\mu}_1 - \boldsymbol{\Sigma}_0^{-1}\boldsymbol{\mu}_0)^T\boldsymbol{x}$, and the constant w_0, where $-\frac{1}{2}\log|\boldsymbol{\Sigma}_1| + \frac{1}{2}\log|\boldsymbol{\Sigma}_0| = \frac{1}{2}\log\big(|\boldsymbol{\Sigma}_0|/|\boldsymbol{\Sigma}_1|\big)$. □

The decision boundary $a = 0$ of naive Bayes is therefore quadratic in $\boldsymbol{x}$, unlike the hyperplane of LDA; only for $\boldsymbol{\Sigma}_0 = \boldsymbol{\Sigma}_1$ does the quadratic term vanish ($\boldsymbol{W}_2 = \boldsymbol{0}$), leaving a linear boundary. Figure 25 compares them for three classes with equal priors and means $(-1, 1)$, $(-1, -1)$, $(1.5, 0)$: naive Bayes uses the covariances $\operatorname{diag}(0.49, 0.16)$ for the first two classes and $\operatorname{diag}(0.16, 0.5625)$ for the third, LDA their average. The first two classes share their covariance in both models, so the boundary between them is a straight line.

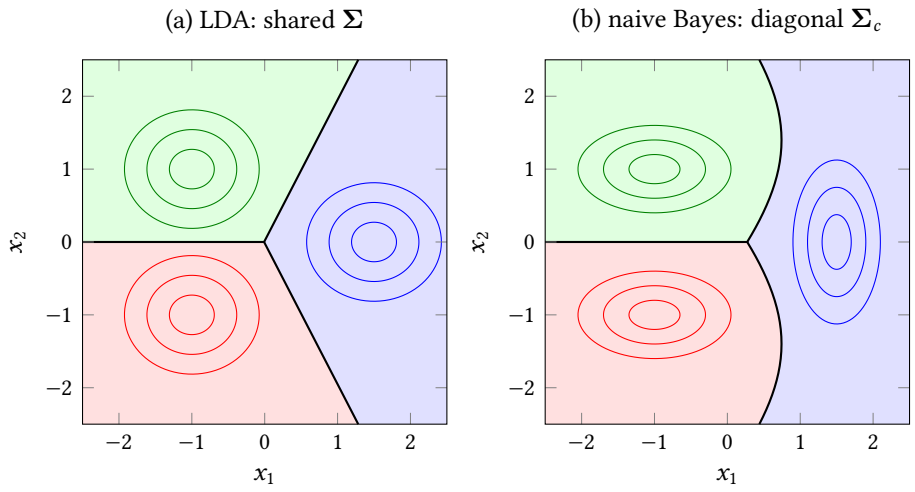


Figure 25: Decision boundaries of LDA and Gaussian naive Bayes.

Because the class conditional factorises, each feature can have its own univariate distribution, e.g. a Gaussian for a continuous feature x_1 and a categorical distribution for a discrete feature x_2: naive Bayes easily handles mixed data types.

5.3 Discriminative models: logistic regression

From LDA to logistic regression. LDA gives $p(y = 1 \mid \boldsymbol{x}) = \sigma(\boldsymbol{w}^T\boldsymbol{x} + w_0)$ and $p(y = 0 \mid \boldsymbol{x}) = 1 - \sigma(\boldsymbol{w}^T\boldsymbol{x} + w_0)$, where $\boldsymbol{w}$ and w_0 are determined by $\boldsymbol{\mu}_0$, $\boldsymbol{\mu}_1$, $\boldsymbol{\Sigma}$ and the class priors (proposition 5.10). Logistic regression keeps this form of the posterior and treats $\boldsymbol{w}$ and w_0 as free parameters, chosen directly.

Definition 5.15 (Logistic regression)**. Logistic regression** models the posterior as

$$y \mid \boldsymbol{x} \sim \mathrm{Ber}\big(\sigma(\boldsymbol{w}^T\boldsymbol{x} + w_0)\big), \qquad \sigma(a) = \frac{1}{1 + \exp(-a)},$$

with free parameters $\boldsymbol{w}$ and w_0. As in definition 4.1, the bias is absorbed by setting $x_0 = 1$, i.e. $\boldsymbol{w}^T\boldsymbol{x} := w_0 + w_1x_1 + \cdots + w_Dx_D$, so that $p(y = 1 \mid \boldsymbol{x}, \boldsymbol{w}) = \sigma(\boldsymbol{w}^T\boldsymbol{x})$.

Maximum likelihood and cross entropy. Learning means finding weights $\boldsymbol{w}$ that explain the training set $\mathcal{D} = \{(\boldsymbol{x}_i, y_i)\}_{i=1}^N$. If the samples are i.i.d., the likelihood is

$$\begin{aligned} p(\boldsymbol{y} \mid \boldsymbol{w}, \boldsymbol{X}) &= \prod_{i=1}^N p(y_i \mid \boldsymbol{x}_i, \boldsymbol{w}) = \prod_{i=1}^N p(y = 1 \mid \boldsymbol{x}_i, \boldsymbol{w})^{y_i}\big(1 - p(y = 1 \mid \boldsymbol{x}_i, \boldsymbol{w})\big)^{1-y_i} \\ &= \prod_{i=1}^N \sigma(\boldsymbol{w}^T\boldsymbol{x}_i)^{y_i}\big(1 - \sigma(\boldsymbol{w}^T\boldsymbol{x}_i)\big)^{1-y_i}. \end{aligned}$$

The second step writes the Bernoulli probability as for the coin (section 3): for $y_i = 1$ the second factor is 1, for $y_i = 0$ the first.

Definition 5.16 (Binary cross entropy)**.** The negative log-likelihood of logistic regression,

$$E(\boldsymbol{w}) = -\log p(\boldsymbol{y} \mid \boldsymbol{w}, \boldsymbol{X}) = -\sum_{i=1}^N \Big(y_i \log \sigma(\boldsymbol{w}^T\boldsymbol{x}_i) + (1 - y_i)\log\big(1 - \sigma(\boldsymbol{w}^T\boldsymbol{x}_i)\big)\Big), \tag{26}$$

is the **binary cross entropy**. The maximum likelihood estimate of $\boldsymbol{w}$ is $\boldsymbol{w}^* = \arg\min_{\boldsymbol{w}} E(\boldsymbol{w})$.

This problem has no **closed-form solution**: the optimal $\boldsymbol{w}^*$ cannot be written with standard operations such as products and matrix inversions. It is solved numerically by optimisation, the topic of the next lecture; for now $\boldsymbol{w}^*$ is assumed to be available.

Regularisation. As seen for linear regression, maximum likelihood estimation often overfits. Penalising large weights controls this:

$$E(\boldsymbol{w}) = -\log p(\boldsymbol{y} \mid \boldsymbol{w}, \boldsymbol{X}) + \lambda\|\boldsymbol{w}\|_q^q, \qquad \|\boldsymbol{w}\|_q^q = \sum_j |w_j|^q.$$

For $q = 2$ this corresponds, as for ridge regression (proposition 4.10), to MAP estimation with a Gaussian prior on $\boldsymbol{w}$. Again there is no closed-form solution.

Multiclass logistic regression. For two classes the sigmoid squeezes the score $\boldsymbol{w}^T\boldsymbol{x} \in \mathbb{R}$ into $(0, 1)$; for C classes the softmax (definition 5.11) does the same with one weight vector per class:

$$p(y = c \mid \boldsymbol{x}, \boldsymbol{w}) = \frac{\exp(\boldsymbol{w}_c^T\boldsymbol{x})}{\sum_{c'=1}^C \exp(\boldsymbol{w}_{c'}^T\boldsymbol{x})},$$

where $\boldsymbol{w}$ collects $\boldsymbol{w}_1, \ldots, \boldsymbol{w}_C$. The labels are encoded by **one-hot encoding**: the label vector $\boldsymbol{y} \in C^N$ becomes the binary matrix $\boldsymbol{Y} \in \{0, 1\}^{N \times C}$ with

$$y_{ic} = \mathbb{I}(y_i = c) = \begin{cases} 1 & \text{if sample } i \text{ belongs to class } c, \\ 0 & \text{otherwise.} \end{cases}$$

As for the categorical prior, $p(y_i \mid \boldsymbol{x}_i, \boldsymbol{w}) = \prod_c p(y_i = c \mid \boldsymbol{x}_i, \boldsymbol{w})^{y_{ic}}$.

Definition 5.17 (Cross entropy)**.** The negative log-likelihood of multiclass logistic regression,

$$E(\boldsymbol{w}) = -\log p(\boldsymbol{Y} \mid \boldsymbol{w}, \boldsymbol{X}) = -\sum_{i=1}^{N}\sum_{c=1}^{C} y_{ic} \log p(y_i = c \mid \boldsymbol{x}_i, \boldsymbol{w}) = -\sum_{i=1}^{N}\sum_{c=1}^{C} y_{ic} \log \frac{\exp(\boldsymbol{w}_c^T \boldsymbol{x}_i)}{\sum_{c'} \exp(\boldsymbol{w}_{c'}^T \boldsymbol{x}_i)},$$

is the **cross entropy**.

Generative versus discriminative models. Table 5 compares the two approaches.

	Generative	Discriminative
Models	$p(\boldsymbol{x} \mid y = c)$ and $p(y = c)$	$p(y = c \mid \boldsymbol{x})$
Strengths	handle missing data better, detect outliers, generate new data, more appropriate in the semi-supervised setting	in general better performance on pure classification tasks
Weaknesses	work reasonably well when their assumptions hold, but are fragile when they are violated; modelling high-dimensional or strongly correlated data (images, graphs) is still an open research problem	

Table 5: Generative and discriminative classifiers.